



### Why do we need this GenC learning Program?

Gen C learning program engages young talents with a comprehensive learning pathway, giving the millennials an opportunity to interact with Subject Matter Experts (SME), understand the corporate environment, and groom themselves.

Cognizant emphasizes on Learner Autonomy where students take charge of their own learning, with the available tools and resources. More focus is on “learning” than “teaching”. Get ready to embark your own learning adventure!

### Program at a glance

Learning consisting of 2 Stages:

- Stage 1 – Core Java with SQL
- Stage 2 – Functional Testing With Selenium Automation Concepts – Inclusive of the Project Business Aligned Project

Interim Project Evaluation + Technical Evaluation

Final Project Evaluation + Final Technical Evaluation

### Program Highlights

- The complete learning journey is formalized using adult learning principles, where problem solving and applying the skills gained are given more importance than conceptual learning.
- Learner Autonomy is encouraged via Flipped Classroom, where the learning platform offers world class learning resources, and students would not be constrained by tutelage of an instructor.
- Get mentored by SME, whose motivation and guidance will help you accelerate in the learning journey.
- This program is applicable to Interns as well as GEN Cs.

# Know Your Service Line – QEA (Quality Engineer and Assurance)

Cognizant Quality Engineering & Assurance (QEA) focuses on ensuring the quality and reliability of enterprise processes, applications, and systems. QEA offers a comprehensive range of services including intelligent and automated quality assurance, modernization assurance, and experience assurance. These services are designed to accelerate business and technology changes, improve customer experiences, and ensure regulatory compliance. QEA also emphasizes the importance of automation and AI to enhance testing efficiency and deliver high-quality outcomes. By leveraging these advanced technologies, QEA helps businesses achieve digital success and maintain competitive advantage in the evolving digital landscape.

## **How QEA Transforms Quality Assurance at Cognizant?**

Cognizant's Quality Engineering & Assurance (QEA) significantly transforms quality assurance by accelerating digital transformation and ensuring robust, reliable applications and systems. This seamless transition minimizes disruptions and speeds up technology adoption.

By embedding quality at every development stage, QEA enhances customer satisfaction and loyalty through high-performance, user-friendly products. Automation and AI-driven quality assurance streamline processes, reducing manual efforts and speeding up release cycles, which boosts operational efficiency and cuts costs. Rigorous testing and validation ensure compliance with industry regulations, crucial for sectors like healthcare and finance.

QEA also supports innovation by providing a reliable foundation for launching new products and entering new markets. Additionally, automated testing frees employees to focus on strategic tasks, enhancing productivity and job satisfaction. These combined impacts help businesses achieve strategic goals, maintain a competitive edge, and deliver exceptional value to their customers.

## **Notable & Successful Stories on Quality Assurance by QEA Cognizant Team...**

Cognizant's Quality Engineering & Assurance (QEA) team partnered with a major health plan provider to overhaul their software quality assurance processes, addressing several critical challenges. The existing methods were outdated and inefficient, struggling to keep pace with the healthcare industry's evolving needs.

Ensuring compliance with stringent healthcare regulations was a significant concern, and the manual, fragmented processes led to delays and increased operational costs. Additionally, there was a pressing need to improve software quality to enhance patient care and overall service delivery.

QEA implemented automated testing frameworks and AI-driven quality assurance tools, significantly reducing manual efforts and accelerating the testing process. Rigorous testing and validation processes were established to ensure compliance with healthcare regulations, including continuous monitoring and real-time compliance checks.

By streamlining quality assurance processes and integrating advanced technologies, QEA improved the overall efficiency of the software development lifecycle.

The focus on quality from the start ensured that end products were reliable and user-friendly, leading to better patient care and satisfaction. As a result, the health plan provider saw a significant improvement in software quality and compliance, which was crucial for meeting healthcare regulations.

The enhancements in software quality directly contributed to better patient care and service delivery, while the automation and streamlined processes led to faster release cycles and reduced operational costs.

The time-to-market for software updates was significantly reduced, allowing the client to respond more quickly to market demands. This project exemplifies how Cognizant's QEA services can transform quality assurance processes, ensuring high-quality outcomes.

[Health plan rebuilds software QA with Cognizant Case Study | Cognizant](#)

## **Tips for Successfully Carrying Best Practices for Implementing and Maintaining Robust IT Infrastructure with QEA.**

**Understand the Basics:** Start with a solid understanding of software development and testing fundamentals. Familiarize yourself with key concepts such as the software development lifecycle (SDLC), different types of testing (e.g., unit, integration, system, and acceptance testing), and the importance of quality assurance.

**Learn Automation Tools:** Gain proficiency in popular automation tools like Selenium, JUnit, and TestNG. Automation is crucial for efficient testing, and being skilled in these tools will enhance your ability to create reliable and repeatable tests.

**Adopt the Page Object Model (POM):** Use the Page Object Model design pattern to create maintainable and reusable code. This approach separates test logic from page elements, making your tests more organized and easier to manage.

**Focus on Continuous Integration/Continuous Deployment (CI/CD):** Understand and implement CI/CD practices. Tools like Jenkins help integrate testing into the development pipeline, ensuring that code changes are tested continuously and deployed seamlessly.

**Embrace Agile and DevOps Practices:** Agile and DevOps methodologies promote collaboration, speed, and quality. Learn how to work in these environments, focusing on continuous improvement and adaptability.

**Prioritize Test Coverage:** Develop comprehensive test plans that cover all aspects of the application, including functional, performance, security, and usability testing. Ensure that your tests are thorough and cover various scenarios.

**Leverage AI and Machine Learning:** Explore how AI and machine learning can enhance testing processes. These technologies can help predict potential issues, optimize test cases, and improve the overall effectiveness of quality assurance.

**Implement Data-Driven Testing:** Use data-driven testing to run the same test with different sets of data. This approach increases test coverage and ensures that your application works correctly with various inputs.

**Maximize Browser Window:** Ensure that your tests maximize the browser window to capture full-page screenshots and interact with all elements. This practice helps improve test accuracy and reliability.

**Validate with Assertions:** Use assertions to validate the outcomes of your tests. Assertions help ensure that the application behaves as expected and quickly identify any issues.

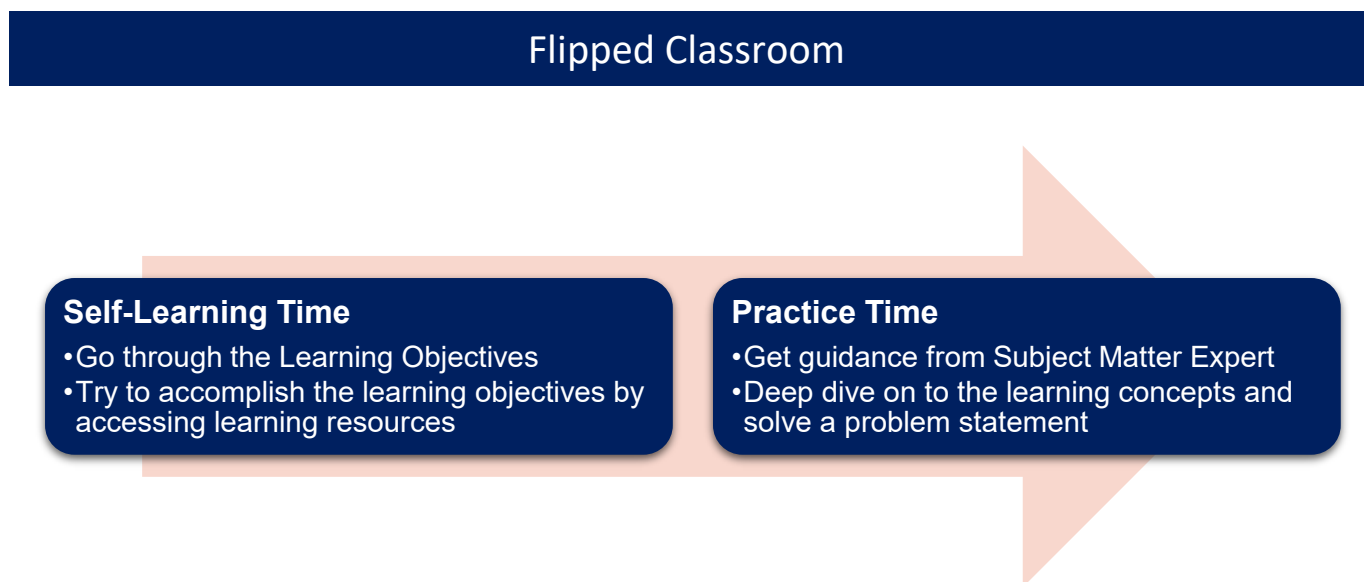
**Continuous Learning and Improvement:** Stay updated with the latest trends, tools, and best practices in quality assurance. Participate in training sessions, workshops, and online courses to continuously enhance your skills.

**Collaborate and Communicate:** Foster strong communication and collaboration with development, testing, and operations teams. Quality assurance is a shared responsibility, and effective teamwork is essential for success.

## Learning Journey with Flipped Classroom

This program encourages you to be more autonomous learners during guided self-learning hours, completing the learning objectives on your own pace and style, and get ready for the hands-on practice time.

The complete learning path is set in the [GEN C Learn Platform](#), which you can login with SSO.

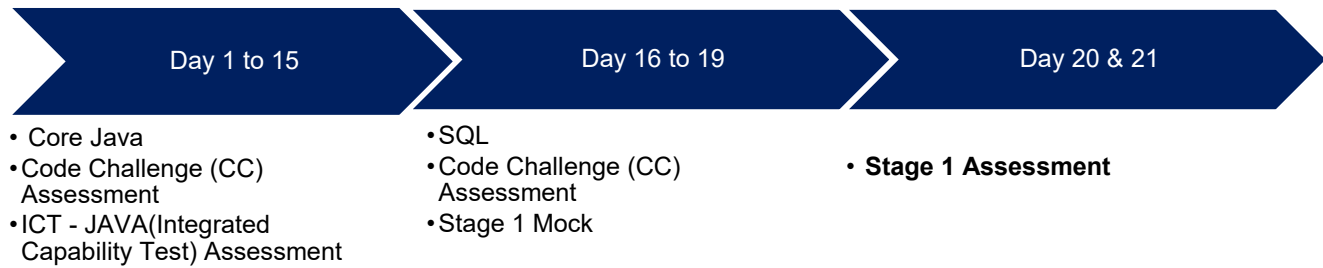


## Recommended Program Sequence

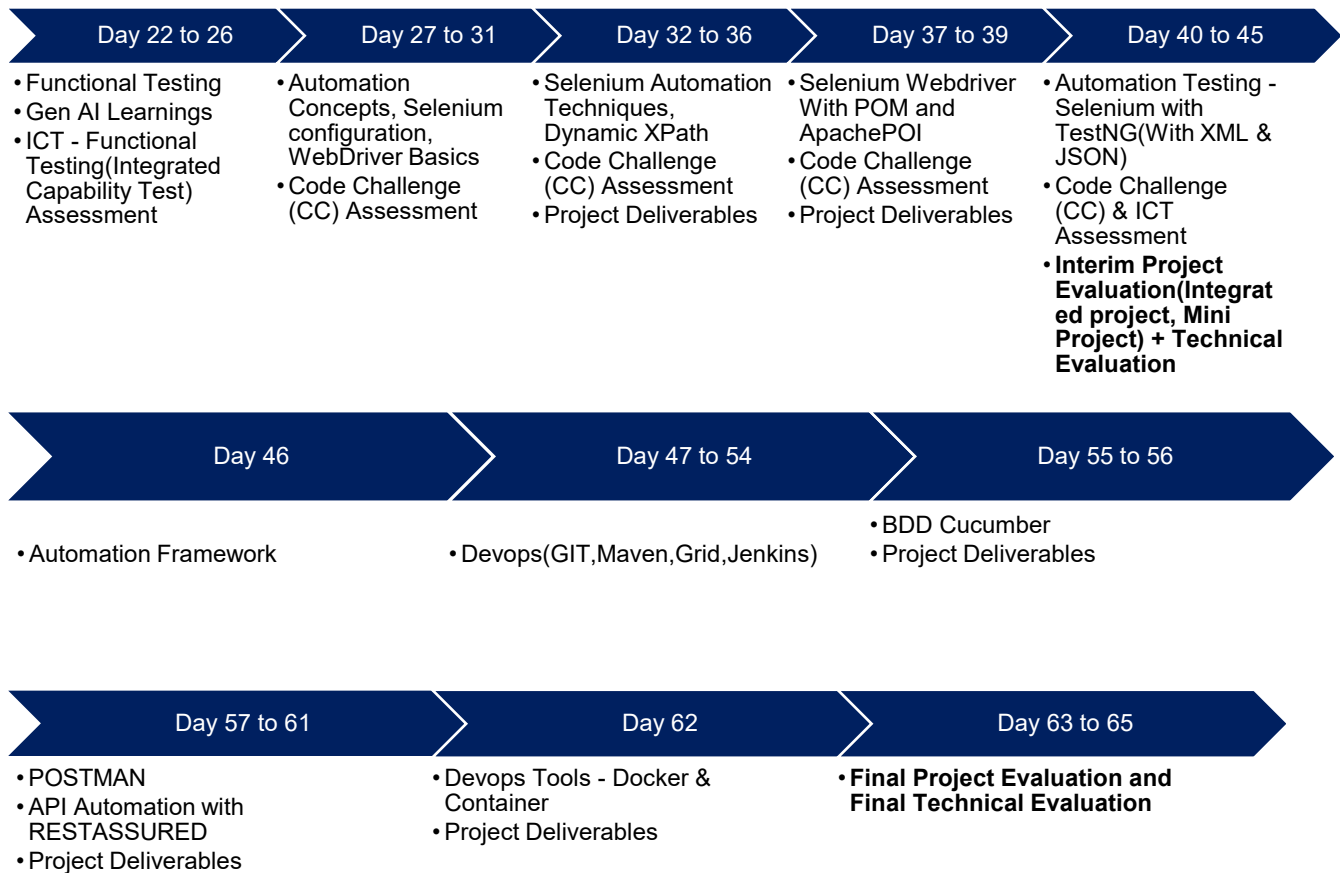
The learning journey contains 2 stages, followed by a Business Aligned Project.

- Stage 1 – Core Java with SQL
- Stage 2 – Functional Testing with Selenium Automation Concepts
- Business Aligned Project will provide you an experience of real time problem solving in Agile methodology.

## Stage 1 – Core Java with SQL



## Stage 2 – Functional Testing with Selenium Automation Concepts



## Key Learning Components of the Program

Cognizant has collaborated with Udemy to provide world class learning videos for the evolving future of work. These Udemy programs are woven into a learning path, empowering you to plan and learn at your style.

The program also connects you with Subject Matter Experts (SMEs) to get the professional guidance on your queries in the learning journey.

The program doesn't ONLY concentrate on the technical skilling, but also on the shaping up of the Behavioral skills. **Behavioral learning** would be done in ILT mode, with few Self-paced learning modules too.

### Evaluation Model

The program continuously evaluates if you can apply those self-learnt skills to solve a real-time business problem. Depicted below are the four key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

- Interim Evaluation (Project + Technical) through Video Interview
- Final Evaluation (Project + Technical) through Video Interview

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC. Additional Learning Components like Hands-On, Code Challenges and ICTs will help you to enhance your expertise level.

## Interim & Final Evaluation Approach

Below is the Evaluation Structure for GenC Learning Journey

The interim evaluation will be held halfway through the learning journey, while the final evaluation will take place at the end of the learning journey.

During the interim evaluation, the GenC will be interviewed by a Technical Subject Matter Expert (SME) from the Business Unit (BU) to assess your knowledge through a technical discussion. Additionally, the Mini project completed by the GenC will also be evaluated by the BU SME. Please note that the Mini project is an individual activity and not a group activity.

During the final evaluation, the GenC will again be interviewed by a Technical SME from the BU to assess their knowledge through a technical discussion. In the same evaluation call, the Hackathon project completed by each group will be evaluated by the BU SME. Please note that the Hackathon project is a group activity and not an individual activity.

## Preparatory Learning

For this automation track training, here are some preparatory learning materials that can be helpful for the interim and final technical evaluations. Preparatory learning material refers to resources, materials, and content that you can use to prepare. These materials are designed to provide you with the necessary knowledge, skills, and understanding to succeed in your learning objectives.

Remember, apart from these learning materials, actively participating in the training sessions, asking question and practicing hands-on exercises will greatly contribute to your overall learning and preparedness for the evaluations.

## Program Completion Criteria

Stage 1 (Pre-requisites)

Gating Criteria: Qualifier Assessment

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| <b>Stage 2 &amp; Beyond (Advanced Skills)</b><br><b>Gating Criteria: Performance Health Status is Green</b> |                                          |                    |                    |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------|--------------------|
| GenC/Intern Training                                                                                        | Evaluation Components                    | Pass Criteria      | Evaluation Done by |
| <b>Performance Health Status - PHS (only from Stage 2)</b>                                                  | Interim Evaluation (Project + Technical) | Green, 1 Attempt   | BU SME             |
|                                                                                                             | Final Evaluation (Project + Technical)   | Green, 2 Attempts* |                    |

100% Completion of Hands On in Stage 2 is mandatory for interim / final evaluation eligibility.

## Key Check Point Intervals in the Learning Journey

### RAG as PHS (Performance Health Status)

The program continuously evaluates if you are able to apply those self-learnt skills to solve a real time business problem. Depicted below are the two key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

#### Interim Evaluation:

- During the interim evaluation, the GenC will undergo a video interview on the learning platform.
- This interview will be conducted by a Tech SME from the BU.
- The purpose of this evaluation is to assess the GenC's knowledge and understanding of the skills covered in the training program up to the halfway point.
- It also encompasses an evaluation of the GenC's progress in their Project case study.
- The evaluation will involve a technical discussion as well as an assessment of the Project progression to gauge the GenC's proficiency in the skills learned thus far.
- There will be ONLY one attempt.

#### Final Evaluation:

- For the final evaluation, the GenC will participate in a video interview conducted by a Tech SME from the BU.
- This evaluation aims to assess the GenC's knowledge and expertise in all the skills covered throughout the entire training program.
- Similar to the interim evaluation, this assessment will involve a technical discussion via a video interview on the learning platform, along with a project evaluation to assess the GenC's capabilities and their Project case study progress.
- It serves as a comprehensive evaluation of the GenC's skills and capabilities acquired during the training.
- In Final Evaluation, Other 1 attempt is given when student PHS is AMBER/RED. (Attempt 1 is not applicable if the student is **Red** in Interim and Final Evaluation - Attempt 1)

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC. Additional Learning Components like Hands-On, Quizzes, CCs, and ICTs will help you to enhance your expertise level.

### **Mandatory Hands-On Exercise Completion :**

- Completion of 100% of the hands-on exercises is mandatory to qualify for the stage 1 qualifier, interim, and final evaluations.

**Table - Check Point Intervals**

| Check points              | Interpreting Status                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interim Evaluation</b> | - <b>Green</b> - On Track for Graduation<br>- <b>Red /Amber</b> - There will not be any re-attempts given                                                                                                                                                                                                                 |
| <b>Final Evaluation</b>   | - <b>Green</b> - On Track for Graduation<br>- <b>Red /Amber</b> – Only 2 attempts are given<br>- (Attempt 2 is not applicable if the student is <b>Red</b> in Interim and Final Evaluation- Attempt 1)<br><br><b>Note:</b> If student fails after the applicable re-attempts, they will be considered as “Not-Graduated”. |

## Icebreaker



Icebreaker session will be conducted for a duration of initial **5 days**. During the session, various topics related to Corporate Induction, Talent Management, Cognizant Agenda on Core Values, Leader Talks, Alumni, BU Mentor connects will be covered. Followed by icebreaker, technical training will kick start.

Following sessions will be covered during the **5 days** of icebreaker

- Corporate Induction
- Talent Manager Connect
- Cognizant Agenda Sessions on Core Values
- Leader Talks (Academy) and many more...

A recommended day-wise schedule is provided below for the learning, with the learning content for the day, the practice hands-on and extended hands-on to be done for the day or

any other activities are listed. Few days might be interleaved to accommodate the extension due to Behavioral Training.

**Mock Interview Session:** This session is conducted to test candidate understanding and enhance candidates' best preparation for Interim & Final Evaluations.

## GenAI-Program Overview

### Introduction

AI Accelerate is a comprehensive program designed to empower learners with the knowledge and skills needed to harness the transformative potential of Artificial Intelligence (AI). This handbook serves as a guide, providing essential information and resources to help learners navigate through the program successfully. From understanding the fundamentals of AI to apply the advanced concepts in real-world scenarios. AI Accelerate is your gateway to unlock the power of AI and shaping the future of technology.

### Program overview.

AI Accelerate offers a learning opportunity, allowing GenCs to engage learning through the self-paced learning, Expert connect, and Knowledge assessment to measure the skills.

### Focus areas

- **Learning:** GenCs will have access to curated content and resources that cover a wide range of topics related to AI, including best practices, and case studies. This learning aspect aims to deepen GenCs understanding of AI and its applications in various industries.
- **Expert connect:** GenCs will have the opportunity to connect with expert in the field of AI. This expert will provide guidance, support, and insights to help GenCs navigate their learning journey and gain valuable insights into the industry.
- **Practice sessions:** GenCs can practice the use cases provided sessions that are designed to reinforce their learning and help them apply their knowledge in real-world scenarios. These sessions will provide GenCs with hands-on experience and practical skills that are essential for success in the field of AI.

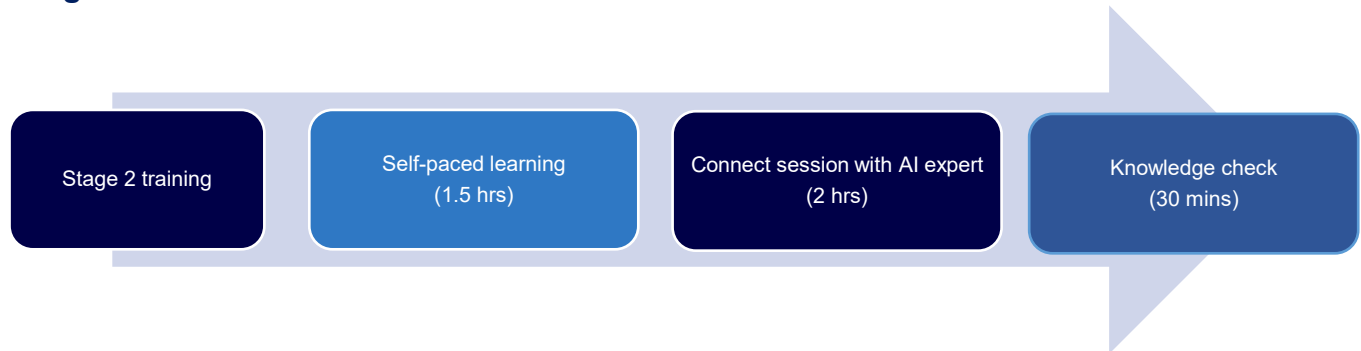
### Performance outcomes

Upon completing the self-paced learning component of AI Accelerate GenCs are expected to achieve the following performance outcomes:

- GenCs will demonstrate a thorough understanding of the fundamentals of AI, including key concepts, terminology, and principles.
- GenCs will be able to apply AI concepts and techniques to solve real-world problems, demonstrating their ability to analyze, design, and implement AI solutions.

- GenCs will develop and apply critical thinking and problem-solving skills in the context of AI, enabling them to identify, analyze, and address complex challenges.
- GenCs will collaborate effectively with peers, mentors, and industry experts to achieve common goals and contribute to the advancement of AI knowledge and practice.

### Program workflow



## Java Programming - Overview

### What is Java Programming?

**Java Programming** is a versatile and widely used programming language that enables developers to create robust, high-performance applications. The process of Java programming involves several stages to ensure the development of efficient and maintainable software.

### Purpose

Java programming is designed to be platform-independent, allowing developers to write code that runs on any device with a Java Virtual Machine (JVM). It emphasizes object-oriented principles, making it easier to create modular, reusable, and maintainable code.

**For Example, Mobile Applications:** Java is the primary language for Android app development. Many popular apps like Spotify, Twitter, and Netflix are built using Java12.

**Enterprise Applications:** Java is extensively used in enterprise environments for building large-scale systems such as banking applications, customer relationship management (CRM) tools, and enterprise resource planning (ERP) solutions.

### Stages of Java Programming:

1. **Requirement Analysis:** Understand and gather the requirements for the Java application. This involves discussions with stakeholders to define the scope and objectives.

2. **Design:** Create a blueprint for the application. This includes designing the architecture, selecting design patterns, and planning the user interface and data flow.
3. **Development:** Write the Java code based on the design specifications. This stage involves coding, unit testing, and integrating different modules.
4. **Testing:** Perform various tests (unit, integration, system, and acceptance testing) to ensure the application functions correctly and meets the requirements.
5. **Deployment:** Deploy the Java application to the production environment. This may involve setting up servers, configuring environments, and ensuring the application is accessible to users.
6. **Maintenance:** Provide ongoing support and updates to the application. This includes bug fixes, performance improvements, and adding new features.



## Software Requirements

Below are the software's need to be installed in system from Company Portal, before start proceeding with selenium Handson and projects like mini project and hackathon.

1. Eclipse Org Eclipse IDE 2023-12
2. Azul JDK Version 17
3. Google Project Hosting Selenium Java 4.15.0
4. Google Project Hosting Selenium Server Standalone 4.8.0
5. Apache Software Foundation, The Apache POI 5.2.4
6. Eclipse Org TestNG for Eclipse 7.4.0
7. Jenkins project Jenkins 2.251

NOTE: If genc's are facing any challenges while installation, trainer/mentor will help to install the software.

### Objective to use the software:

- Developing the project

- Writing the code
- Debugging the errors
- Testing the applications

## BU Mandatory - Technical learning and Handson – Core Java

NOTE: The Below topics will be covered by Trainers in Master sessions and relevant

hands-on exercises will be provided as Daily Learning Exercises (DLE) for associates to practice.

Associates must complete the provided Udemy courses along with these BU Connect Master sessions. Hands-on practice and doubt clarification can be addressed during the trainer-led Master sessions.

Associates must also complete the self-paced learning and hands-on exercises available on the Tekstac Platform in parallel with the trainer-led sessions and DLE.

Click Here to View the [BU Aligned Technical Training Contents](#) Document.

## Schedule – Stage 1: Day 1 to 15

Day 1 to 15 will be focusing on Core Java.

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

### Continuous Learning: Technical Hands-on through Tekstac

Post Completion of the ILT session, Trainee must Practice the Handson provided in the Tekstac Platform . In build code editor is available in this platform and Handson are automatically evaluated when the code is compiled in the editor.

### Continuous Learning: Java Practice Hands-on in Eclipse

You will Parallel Learn and complete the Tekstac Platform Hands-on exercises and practice Topic wise Java Practice Handson Daily . This Exercise is set for practice only and not mandate . But completion of Tekstac Platform Handson is mandate for enablement of Assessments.

Follow these steps to practice - Topic Wise Java Practice Hands-on :

1. Software: Download the Eclipse software.
2. Practice Topic-wise Hands-on Exercises: Practice the given topic-wise hands-on exercises in your local environment using Eclipse daily post completion of Learning and platform driven Handson Exercise.
3. Submit for Evaluation: Once you have completed the exercises, submit them for evaluation.

## Day 1

### Core Java

#### Continuous Learning: ILT Enablement

| Chapter   | Chapter Name                                   | Topic                                                                                                                                                                                                                                                                        | Training Mode                                              |
|-----------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Chapter-1 | Introduction to Java, Data Types and Variables | <b>History &amp; Features of Java:</b> OOP, platform independence, JVM, JDK, JRE<br><b>Data Types:</b> Primitive and non-primitive types, Variable declaration, Type casting                                                                                                 | In-Person Trainer - ILT<br>(ILT - Instructor Led Training) |
| Chapter-2 | OOP Basics                                     | <b>Object-Oriented Principles:</b> Encapsulation, Inheritance, Polymorphism, Abstraction<br><b>Classes and Objects:</b> Class structure, Object creation, Instance variables<br><b>Constructors:</b> Default constructor, Parameterized constructor, Constructor overloading |                                                            |
|           |                                                | <b>HandsON : Day 1 Topics - TekStac Lab (Practice In TekStac Platform)</b>                                                                                                                                                                                                   |                                                            |
|           |                                                |                                                                                                                                                                                                                                                                              | Daily Lab Exercise(DLE)                                    |

#### Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

#### Continuous Learning: Technical Hands-on through Tekstac Mandatory Hands-on

- Streamline Your Health Journey
- Water Tank Capacity
- Square Series

#### Continuous Learning: Java Practice Hands-on in Eclipse

##### Basic Syntax

- Simple Java program
- Print an integer value
- Command line arguments
- Print the input from scanner

##### Object Oriented Programming

- Object Oriented Programming

- Constructors
- Abstract Classes
- Interfaces
- Static Members
- Final Keyword
- Miscellaneous
- Factorial of a number using recursion
- Find the factorial of a given number
- See the output as 0, 1, 1, 2, 3
- Find the biggest number among 1,2,3,4,5,65,76,5,4,33,4,34,232,3,2323
- Read and write a file
- Print the below output
- Print the triangle of numbers

## Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks.](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

### Section 2: Java Programming

## Day 2

## Core Java

### Continuous Learning: ILT Enablement

| Chapter   | Chapter Name                             | Topic                                                                                   | Training Mode           |
|-----------|------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------|
| Chapter-3 | Inheritance, Abstraction & Encapsulation | <b>Inheritance:</b> Single inheritance, Multilevel inheritance, super keyword           | In-Person Trainer - ILT |
|           |                                          | <b>Polymorphism:</b> Method overloading, Method overriding                              |                         |
|           |                                          | <b>Abstraction:</b> Abstract classes, Interfaces                                        |                         |
|           |                                          | <b>Encapsulation:</b> Access modifiers (public, private, protected), Encapsulating data |                         |
|           |                                          | <b>HandsON : Day 2 Topics - TekStac Lab (Practice In TekStac Platform)</b>              | DLE & BU Connects       |

### Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

### Continuous Learning: Technical Hands-on through Tekstac Mandatory Hands-on

- WordZee
- Fishing competition
- Brilliant Restaurant
- Investment Calculation

## Continuous Learning: Java Practice Hands-on in Eclipse

### Basic Arithmetic Operations

- Java program to swap two numbers
- Convert from Fahrenheit to Celsius
- Swap two numbers without using a third variable
- Java program to add two numbers
- Java program to find the GCD of two numbers
- Java program to find the LCM of two numbers
- Java program to print the sum of digits of a number
- Java program to find the GCD of two numbers
- Java program to find the LCM of two numbers
- Java program to print the sum of digits of a number

## Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

### Section 2: Java Programming

Day 3

## Core Java

## Continuous Learning: ILT Enablement

| Chapter   | Chapter Name                                | Topic                                                                                        | Training Mode           |
|-----------|---------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------|
| Chapter-4 | Keywords                                    | <b>this and super:</b> Referring current object, Accessing parent class methods              | In-Person Trainer - ILT |
|           |                                             | <b>Static Members:</b> Static variables, Static methods, Static blocks                       |                         |
|           |                                             | <b>final Keyword:</b> Final variables, Final methods, Final classes                          |                         |
| Chapter-5 | Arrays,Strings,StringBuffer & StringBuilder | <b>Arrays:</b> One-dimensional array, Multidimensional array, Array traversal and operations | DLE & BU Connects       |
|           |                                             | <b>Strings:</b> String class, Immutability, String methods: length, substring, equals, etc.  |                         |
|           |                                             | <b>HandsON : Day 3 Topics - TekStac Lab (Practice In TekStac Platform)</b>                   |                         |

## Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

## Continuous Learning: Technical Hands-on through Tekstac

### Mandatory Hands-on

- Dazzle Closet
- Mass Of Rocket Component
- Lego Builders
- Plan Smart
- Mythic Rhythms

## Continuous Learning: Java Practice Hands-on in Eclipse

### Control Flow Statements

- Demonstrate if...else statement
- Demonstrate nested if...else if... statement
- Demonstrate nested if...else statement
- Find odd and even numbers
- Printing the prime numbers

### Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

### Section 2: Java Programming

## Day 4

## Core Java

## Continuous Learning: ILT Enablement

| Chapter   | Chapter Name                                | Topic                                                                            | Training Mode                                |
|-----------|---------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------|
| Chapter-5 | Arrays,Strings,StringBuffer & StringBuilder | StringBuffer & StringBuilder: Mutability, Common methods: append, insert, delete | In-Person Trainer - ILT<br>DLE & BU Connects |
| Chapter-6 | Exception Handling                          | Types of Exceptions: Checked exceptions, Unchecked exceptions                    |                                              |
|           |                                             | try-catch Block: Single try-catch, Multiple catch, finally block                 |                                              |
|           |                                             | throw and throws: Manually throwing exceptions, Propagating exceptions           |                                              |
|           |                                             | HandsON : Day 4 Topics - TekStac Lab (Practice In TekStac Platform)              |                                              |

## Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

## Continuous Learning: Technical Hands-on through Tekstac

### Mandatory Hands-on

- Fitness Tracker
- Array Aggregate
- Product Details

## Continuous Learning: Java Practice Hands-on in Eclipse

### Loops

- Demonstrate for loop
- Print stars using for loop, where the number of stars printed should be equal to the row number
- Demonstrate while loop
- Print the entered number in reverse
- Demonstrate the usage of break statement inside while loop
- Print the alphabets using for loop
- Demonstrate the usage of break and continue statements inside while loop
- Demonstrate for each loop
- Print Floyd's triangle
- Print Pascal's triangle
- Print the Fibonacci series
- Printing the Multiplication table
- Find the factorial of a number
- Print the prime numbers up to a given number
- Calculate the power of a number using a while loop
- Verify whether a number is a perfect number or not
- Printing the Fibonacci series from 1 to 10
- Find the greatest of three numbers
- Print the largest two numbers from an array of given numbers

### Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

### Section 2: Java Programming

### Additional Learning:

#### Technical Quizzes:

- Java Operator, Control flow statement

Day 5

### Core Java

## Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name       | Topic                                                                                                                       | Training Mode                                |
|-----------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Chapter-6 | Exception Handling | Custom Exceptions: Creating custom exceptions                                                                               | In-Person Trainer - ILT<br>DLE & BU Connects |
| Chapter-7 | Java Collections   | Introduction to Collections: Collection hierarchy, Collection interfaces (List, Set, Map)                                   |                                              |
|           |                    | List Interface: ArrayList, LinkedList, Vector, Stack<br>HandsON : Day 5 Topics - TekStac Lab (Practice In TekStac Platform) |                                              |

## Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

## Continuous Learning: Technical Hands-on through Tekstac

### Mandatory Hands-on

- College Namelist
- Babitha's App
- Alta Motors

## Continuous Learning: Java Practice Hands-on in Eclipse

### Method Overloading and Overriding

- Final methods be overloaded
- Static methods be overloaded
- Final methods be overridden
- Static methods be overridden
- Overload a main method in Java
- Object u have created

## POLYMORPHISM & INHERITANCE

- Demonstrate method overloading
- Demonstrate method overriding
- Demonstrate inheritance
- Demonstrate the use of the 'super' keyword

## Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

### Section 2: Java Programming

Day 6

Core Java

## Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name     | Topic                                                                      | Training Mode                |
|-----------|------------------|----------------------------------------------------------------------------|------------------------------|
| Chapter-7 | Java Collections | <b>Set Interface:</b> HashSet, LinkedHashSet, TreeSet                      | In-Person Trainer - ILT      |
|           |                  | <b>Map Interface:</b> HashMap, LinkedHashMap, TreeMap, Hashtable           |                              |
|           |                  | <b>Iterator:</b> Iterator, ListIterator                                    |                              |
|           |                  | <b>HandsON : Day 6 Topics - TekStac Lab (Practice In TekStac Platform)</b> | <b>DLE &amp; BU Connects</b> |

### Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

### Continuous Learning: Java Practice Hands-on in Eclipse

#### Arrays

- Find the largest number
- Remove duplicates from an array
- Find the second largest number in an array?
- Sort an array using bubble sort
- Sort an array using selection sort
- Sort an array using insertion sort
- Implement binary search
- Implement linear search
- Find the sum of elements in an array
- Find the average of elements in an array
- Find the maximum and minimum elements in an array
- Remove duplicates from an array
- Find the smallest and largest numbers in a Java Array
- Java program to swap two strings without using temp or third variable
- Demonstrate creating an array
- Duplicate elements in an array of numbers
- Demonstrate creating a multidimensional array
- Print the elements of the array in reverse
- Print alternative elements in a String array
- Find the greatest number in an integer array
- Find the least number in an integer array
- Using a for loop with single-dimensional arrays
- Using a for-each loop with single-dimensional arrays

### Learn and Practice



#### [Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author

## Additional Learning:

### Serial and Parallel Sorts in Java

- <https://www.geeksforgeeks.org/serial-sort-vs-parallel-sort-java/>

### Streams and Optionals

- <https://www.geeksforgeeks.org/java-8-stream-tutorial/?ref=lbp>

## Day 7

### Core Java

### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name         | Topic                                                                                                                                                          | Training Mode                                |
|-----------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Chapter-7 | Java Collections     | <b>Comparable &amp; Comparator:</b> Sorting with Comparable, Sorting with Comparator                                                                           | In-Person Trainer - ILT<br>DLE & BU Connects |
| Chapter-8 | Java Thread Handling | <b>Introduction to Threads:</b> Thread vs Process, Thread lifecycle                                                                                            |                                              |
|           |                      | <b>Creating Threads:</b> Extending Thread class, Implementing Runnable interface                                                                               |                                              |
|           |                      | <b>Thread Methods:</b> sleep(), join(), start(), run()                                                                                                         |                                              |
| Chapter-9 | Java File Handling   | <b>Character Streams:</b> FileReader, FileWriter, BufferedReader, BufferedWriter<br><b>HandsON : Day 7 Topics - TekStac Lab (Practice In TekStac Platform)</b> |                                              |

### Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

### Continuous Learning: Technical Hands-on through Tekstac Mandatory Hands-on

- Student Score Info
- Vintage Books Emporium

## Practice Section – Integrated Capability Test

Once completing Udemty course, Learnings, Handson related to Java skill appear for the Practice Assessment. This assessment has been set up in platform to help you to assess and apply the concepts learnt as part of Java skill.

You can take up this assessment as a practice till your qualifier assessment for practice. The purpose of this practice assessment is to prepare you for the actual assessment and enhance your information retention. Regular practice helps reinforce what you've learned, making it easier to recall during the actual assessment.

Assessment Attempts: Unlimited

Assessment Duration: 3 hrs

No No.of Questions appear in each attempt : 3

## Continuous Learning: Java Practice Hands-on in Eclipse String Handling

- Java program to compare two strings
- Java program to reverse a string
- Java program to count the number of vowels and consonants in a string
- Java program to find the frequency of characters in a string
- Java program to reverse a string
- Java program to check if two strings are anagrams
- Java program to count the number of vowels and consonants in a string
- Java program to find the frequency of characters in a string
- A string such that the first character is uppercase, the second is lowercase, and so on the numbers from a string and print only the alphabets
- Reverse the words in a sentence
- The frequency of words in a paragraph

## Learn and Practice



### [Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Day 8

## Core Java

## Continuous Learning: Technical Enablement

| Chapter    | Chapter Name                      | Topic                                                                                          | Training Mode                                |
|------------|-----------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------|
| Chapter-9  | Java File Handling                | <b>File Handling:</b> File class, File reading and writing (FileInputStream, FileOutputStream) | In-Person Trainer - ILT<br>DLE & BU Connects |
| Chapter-10 | JDBC (Java Database Connectivity) | <b>JDBC API Overview:</b> JDBC architecture, Driver Manager, Statement and PreparedStatement   |                                              |
|            |                                   | <b>CRUD Operations:</b> Insert, Update, Delete, Select Queries                                 |                                              |
|            |                                   | <b>Transactions:</b> Commit, Rollback, Savepoints                                              |                                              |
|            |                                   | <b>HandsON : Day 8 Topics - TekStac Lab (Practice In TekStac Platform)</b>                     |                                              |

## Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

## Continuous Learning: Technical Hands-on through Tekstac Mandatory Hands-on

- Animalia
- Customer Record

## Continuous Learning: Java Practice Hands-on in Eclipse Exception Handling

- Demonstrate exception handling using try-catch
- Demonstrate the use of finally block
- Demonstrate the use of throw and throws
- Demonstrate custom exceptions
- Demonstrate the advantage of finally in Exception Handling
- Custom class which is immutable

## Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Day 9

Core Java

## Continuous Learning: ILT Technical Enablement

| Chapter    | Chapter Name           | Topic                                                                      | Training Mode           |
|------------|------------------------|----------------------------------------------------------------------------|-------------------------|
| Chapter-11 | Java 8 Features        | <b>Lambda Expressions:</b> Lambda syntax, Functional interfaces            | In-Person Trainer - ILT |
|            |                        | <b>Stream API:</b> Stream operations (filter, map, reduce)                 |                         |
|            |                        | <b>Optional Class:</b> Avoiding NullPointerException with Optional         |                         |
| Chapter-12 | Generics & Annotations | <b>Generics:</b> Generic classes, methods & wildcards                      | DLE & BU Connects       |
|            |                        | <b>HandsON : Day 9 Topics - TekStac Lab (Practice In TekStac Platform)</b> |                         |

## Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

## Continuous Learning: Technical Hands-on through Tekstac Mandatory Hands-on

- Fit Freak
- Employee Details

## Continuous Learning: Java Practice Hands-on in Eclipse Collections Framework

- Java program to demonstrate the use of collections (ArrayList)
- Java program to demonstrate the use of collections (HashMap)
- Java program to demonstrate the use of collections (HashSet)
- Java program to demonstrate the use of collections (LinkedList)
- Java program to demonstrate the use of collections (TreeSet)
- Java program to demonstrate the use of collections (PriorityQueue)
- Java program to demonstrate the use of collections (Stack)
- Java program to demonstrate the use of collections (Vector)

## Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Day 10

## Core Java

## Continuous Learning: ILT Technical Enablement

| Chapter    | Chapter Name           | Topic                                                                           | Training Mode                                |
|------------|------------------------|---------------------------------------------------------------------------------|----------------------------------------------|
| Chapter-12 | Generics & Annotations | Annotations: Built-in annotations (@Override, @Deprecated) & Custom annotations | In-Person Trainer - ILT<br>DLE & BU Connects |
| Chapter-13 | Design Patterns        | Design Patterns: Singleton, Factory, Observer, Strategy patterns                |                                              |
|            |                        | HandsON : Day 10 Topics - TekStac Lab (Practice In TekStac Platform)            |                                              |

## Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

## Continuous Learning: Technical Hands-on through Tekstac Mandatory Hands-on

- Booking System
- Copy Names

## Continuous Learning: Java Practice Hands-on in Eclipse Collections Framework

- Java program to demonstrate the use of collections (Deque)
- Java program to demonstrate the use of collections (LinkedHashMap)
- Java program to demonstrate the use of collections (LinkedHashSet)
- Find duplicates in a list
- Two HashMaps for equality
- Creating an ArrayList
- Creating a LinkedList
- Creating a HashSet
- Creating a LinkedHashSet
- Creating a TreeSet
- Creating a PriorityQueue
- Creating a HashMap using the Map interface
- Creating a LinkedHashMap
- Creating a TreeMap

## Learn and Practice



### [Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

## Continuous Learning: Technical Hands-on

### Additional Learning:

### Technical Quizzes:

- Applying Object Oriented Concepts in java

## Core Java

### Continuous Learning: Technical Enablement through Tekstac

#### Mandatory Learning

Java 8 Features - Streams and Optionals. Asynchronous and Parallel Programming in Java 8. Go through web pages for learning below specific topics

[Asynchronous and Parallel Programming](#)  
[Java User Input \(Scanner\)](#)

### Continuous Learning: Technical Hands-on through Tekstac

#### Mandatory Hands-on

- Mystic Line
- Area And Volume
- The Perfect Password
- Library Management

#### Additional Learning



[Java In-Depth: Become a Complete Java Engineer!](#)

Go through only below mentioned sections 16 & 17 and implement the examples along with the author

- Section 16: Collections Framework (aka Data Structures)
- Section 17: Generics

## JDBC

### Continuous Learning: Technical Enablement Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

#### Additional Learning: Technical Quizzes:

- Quiz 3 - Collections Framework
- Quiz 4 - Advanced Java Concepts

Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Java** skill appear for the **Java -Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

**Assess-Type-1: Code Challenge – Java - Group 1**  
**Assess-Type-1: Code Challenge – Java - Group 2**

Assessment Duration: 2 hrs.  
Assessment Attempts: 3  
No.of Questions appear in each attempt: 1

## Day 13 & 14

### Core Java

#### Continuous Learning: Technical Enablement

##### Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

[Java Database Connection: JDBC and MySQL](#)

- Rehearse the sections based on need

## Day 15

### Core Java

#### Continuous Learning: Technical Enablement

##### Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

[Java Database Connection: JDBC and MySQL](#)

- Rehearse the sections based on need

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Java** skill appear for the **Java - ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

#### **GenC - QAQE-TECHNICAL-JAVA TRACK - CORE JAVA SKILL WAR - SKILL BASED ASSESSMENT [101-BASICS] -LAB\_ATKJE072**

Assessment Duration: 4 hrs.  
Assessment Attempts: 3  
No.of Questions appear in each attempt : 1

## Day 16

### SQL

Day 16 to 21 will be focusing on SQL Language.

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

#### Continuous Learning: Technical Enablement

Learn the basics of SQL



[Relational Database Design](#)

Refer all the sections of the Udemy course

Go through web pages for learning below specific topic- **No SQL overview**

[Introduction to NoSQL](#)



[SQL for Beginners: Learn SQL using MySQL and Database Design](#)

Refer sections 1 to 12, and 14 in this Udemy course and complete the corresponding learnings.

Section 1: Course Introduction

Section 2: Installation and Setup

Section 3: Data Definition Language

Section 4: More on Alter Table

Section 5: Data Manipulation

Section 6: Selecting from a Table

Section 7: Selecting from Multiple Tables

Go through web pages for learning below specific topics

[RANK Function](#)

## Continuous Learning: Technical Hands-on Mandatory Hands-on

- Create Borrowing Table
- Alter Table User
- Alter Table Book
- Remove Table
- Insert User Table
- Update Record
- Delete Record

## Day 17

### SQL

## Continuous Learning: Technical Enablement

Learn the basics of SQL



[Relational Database Design](#)

Refer all the sections of the Udemy course

Go through web pages for learning below specific topic- **No SQL overview**

[Introduction to NoSQL](#)



[SQL for Beginners: Learn SQL using MySQL and Database Design](#)

Refer sections 8, 9, 10, 11 and 12 in this Udemy course and complete the corresponding learnings.

Section 8: Database Design

Section 9: Creating a Cinema Booking System Database

Section 10: Aggregate Functions

Section 11: Subqueries

Section 12: MySQL Functions - String Functions and Date Functions

## Continuous Learning: Technical Hands-on Mandatory Hands-on

- Diverse Selection
- Email Providers
- Book Event
- Upcoming Fair

- Routine Checks
- Security Measure
- May Birthday
- Age Demographics
- Author Detail

### Additional Learning:

#### Technical Quizzes:

Querying Database - Operators, Aggregate, String, Date Functions

### MOCK Stage1 Assessment

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to Java skill appear for the **Mock Stage1 Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assessment Duration: 3 hrs.

Assessment Attempts: 3

No. of Questions: 3

## Day 18

### SQL

#### Continuous Learning: Technical Enablement

Learn the basics of SQL



[Relational Database Design](#)

Refer all the sections of the Udemy course

Go through web pages for learning below specific topic- **No SQL overview**

[Introduction to NoSQL](#)



[SQL for Beginners: Learn SQL using MySQL and Database Design](#)

Refer sections 13 and 14 in this Udemy course and complete the corresponding learnings.

Section 13: Challenges

Section 14: Extra Information -Source code, and other stuff

#### Continuous Learning: Technical Hands-on

#### Mandatory Hands-on

- Price Analysis
- Updated Email
- Lended Books
- The Chronicles of Authors and Their Tales
- Date Format
- Exploring Users' Literary Journeys
- Active Borrowings with User and Book Details
- Same Authors
- User Borrowing Summary
- Maximum Fine Per User
- Top Users
- User Count by Title and Author

### **Additional Learning:**

#### **Technical Quizzes:**

SQL basics with DML and DDL Statements

Once completing Udemey course, Learnings, all Handson & Practice assessments related to **Web UI-SQL** skill appear for the **SQL -Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

#### **Code Challenge**

Assess-Type-1: Code Challenge - DDL, DML & Select (QEA)

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

## **Day 19**

### **SQL**

#### **Continuous Learning: Technical Enablement**

Learn the basics of XML and JSON



[Learn API Technical Writing: JSON and XML for Writers](#)

Refer sections 1, 2, 3 and 4 in this Udemey course and complete the corresponding learnings.

Section 1: Introduction

Section 2: JSON

## Section 3: XML

## Section 4: Final Words

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Web UI-SQL** skill appear for the **SQL -Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

### Code Challenge

Assess-Type-1: Code Challenge - JOINS and SUBQUERIES (QEA)

Assess-Type-1: Code Challenge - Scalar, Aggregate (QEA)

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

## Day 20 & Day 21

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to Java skill appear for the **Stage 1 Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform. Question Pattern - 3 SBA Java Questions approximately spend around 1 hr. for each SBA Question. Score acquiring in 1st instance will be getting into consideration. There are no more attempts provided further.

### Stage1 Assessment

Assessment Duration: 3 hrs.

No. of Questions: 3

## Schedule – Stage 2: Week 22 to 26

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

## Day 22

### Agile Functional Testing

| Topic #  | Topic Name                      | Topic Objective                                               | References             |
|----------|---------------------------------|---------------------------------------------------------------|------------------------|
| Module 1 | Agile Software Testing Concepts | Understanding SDLC and STLC                                   | Theory and Foundations |
|          |                                 | Distinguishing between Project and Product in testing context |                        |
|          |                                 | Exploring V-Model and the roles of QA, QC, and QE             |                        |

|          |                                |                                                                                     |                                                                                                                     |
|----------|--------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|          |                                | Differentiating between White Box and Black Box Testing                             |                                                                                                                     |
|          |                                | Delving into Static Testing and Dynamic Testing                                     |                                                                                                                     |
|          |                                | Conceptualizing Verification and Validation                                         |                                                                                                                     |
|          |                                | System Testing Types (GUI Testing)                                                  |                                                                                                                     |
|          |                                | System Testing Types (Functional vs. Non-Functional Testing)                        |                                                                                                                     |
|          |                                | Test Design Techniques and strategies                                               | Includes Re-Testing, Regression Testing, Exploratory Testing, Adhoc, Sanity, Smoke Testing, and End-To-End Testing. |
|          |                                | Understanding STLC phases (Use case, Test scenario, Test case)                      |                                                                                                                     |
|          |                                | Understanding STLC phases (Test Environment, Execution, Defect Reporting)           |                                                                                                                     |
|          |                                | Understanding STLC phases (Test Closure, and Test Metrics)                          |                                                                                                                     |
| Module 2 | Agile Software Testing Project | Introduction to the Live Project                                                    | Practical Application                                                                                               |
|          |                                | Extracting Functional Requirements from FRS                                         | Requirement Analysis                                                                                                |
|          |                                | Creating comprehensive Test Scenarios and Test Cases                                | Writing Test Artifacts                                                                                              |
|          |                                | Execution of Test Cases                                                             |                                                                                                                     |
|          |                                | Reporting and Tracking Bugs                                                         | Defect Management                                                                                                   |
|          |                                | Test Sign-off                                                                       | Completion Phase                                                                                                    |
| Module 3 | Agile/Scrum Process            | Overview of Agile and Scrum methodologies                                           | Agile Fundamentals                                                                                                  |
|          |                                | Understanding Scrum Teams, Sprints, and User Stories                                |                                                                                                                     |
|          |                                | Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready' |                                                                                                                     |
|          |                                | Exploring Sprint Activities (Planning, Refinement, Review, Retrospective)           |                                                                                                                     |
| Module 4 | JIRA Tool Implementation       | Installation and Configuration of JIRA                                              | Hands-on Tool Setup                                                                                                 |
|          |                                | Project creation in Jira and user management                                        |                                                                                                                     |
|          |                                | Creating Backlog, Epic, and User Stories in JIRA                                    | Requirements in JIRA                                                                                                |
|          |                                | Sprint creation and management                                                      |                                                                                                                     |
|          |                                | Understanding the Sprint life cycle in JIRA                                         |                                                                                                                     |
|          |                                | Bug creation in Jira                                                                | Defect Reporting in JIRA                                                                                            |
|          |                                | Writing test cases in JIRA                                                          | Test Case Management                                                                                                |
|          |                                | Creating Test Cycles and executing/updating Test Cases in Jira                      | Test Execution in JIRA                                                                                              |

## Continuous Learning: Technical Enablement

Learn the basics of Agile Fundamentals and Software Testing Life Cycle



[Learn Manual Software Testing + Agile with Jira Tool](#)

Learn all the sections in this Udemy course to gain deep understanding about below modules

### Module : Software Testing Concepts

- Understanding Software Development Life Cycle (SDLC) and Software Testing Life Cycle (STLC)
- Distinguishing between Project and Product in the context of testing
- Exploring V-Model and the roles of QA (Quality Assurance), QC (Quality Control), and QE (Quality Engineering)
- Differentiating between White Box and Black Box Testing
- Delving into Static Testing and Dynamic Testing
- Conceptualizing Verification and Validation in software testing
- System Testing Types, including GUI Testing and Functional vs. Non-Functional Testing
- Test Design Techniques and strategies like Re-Testing, Regression Testing, Exploratory Testing, Adhoc Testing, Sanity, Smoke Testing, and End-To-End Testing
- Understanding Software Testing Life Cycle (STLC) phases, including Use case, Test scenario, Test case, Test Environment, Execution, Defect Reporting, Test Closure, and Test Metrics.

### Module : Software Testing Project

- Introduction to project
- Extracting Functional Requirements from Functional Requirement Specifications (FRS)
- Creating comprehensive Test Scenarios and Test Cases
- Execution of Test Cases
- Reporting and Tracking Bugs
- Test Sign-off

### Phase : Agile Testing + Jira Tool

#### ***Agile/Scrum Process:***

- Overview of Agile and Scrum methodologies
- Understanding Scrum Teams, Sprints, and User Stories
- Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'
- Exploring Sprint Activities such as Sprint Planning, Backlog Refinement, Sprint Review, and Sprint Retrospective

#### ***Jira Tool:***

- Installation and Configuration of JIRA
- Project creation in Jira and user management
- Creating Backlog, Epic, and User Stories in JIRA
- Sprint creation and management
- Understanding the Sprint life cycle in JIRA
- Bug creation in Jira
- Writing test cases in JIRA with Zephyr plugin
- Creating Test Cycles and executing/updating Test Cases in Jira

## Continuous Learning: Technical Hands-on

### Mandatory Hands-on

- The Importance of QEA and Career Scope
  - The Scope Debate: Quality Assurance vs. Software Development
  - Trainer-Guided Integrated Project: Activity Readiness
  - Trainer-Guided Integrated Project: Configured Jira Project & Backlog Management
  - Trainer-Guided Integrated Project: Test Case Development in Jira (using Zephyr/similar plugin)
  - Trainer-Guided Integrated Project: Test Execution & Status Reporting Artifacts
  - Trainer-Guided Integrated Project: User story document
  - Trainer-Guided Integrated Project: Retrospection Report
  - Trainer-Guided Integrated Project: Sprint Backlog
  - Placeholder to submit other Functional agile deliverables
- ✓ You should have introduced more than one sprints
  - ✓ Ensure you have in-sprint automation concepts introduced as part of project
  - ✓ Ensure you have created Regression suite and smoke suite
  - ✓ Ensure your regression suite consists of minimum 25 test cases and test scripts
  - ✓ Ensure your smoke suite covers end to end business functionality

**Additional Learning:**  
**Technical Quizzes:**  
 Functional Testing

## Day 26

### Functional Testing

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Functional Testing** skill appear for the **Functional Testing -ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform. Attempts allowed to practice the assessment is 3.

Assessment Duration: 4 hrs

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

- Functional Testing -Assess Type 2 Skill Based Assessment.
- Read the requirements carefully of the Assessment Question.
- Write the Defect Description for the Test Case in Excel Sheet
- Submit the Test deliverables

## GenAI (Day 24)

### Continuous Learning: Technical Enablement

#### AI Accelerate

**By this course you will complete GenAI self-pace learning , AI expert connect session and AI Knowledge check assessment.**

#### Program completion criteria

Everyone must register for the following e-learning course on [C-Learn](#) and complete a KBA assessment on Moodle to successfully finish this program.

#### Online learning: C-Learn



Activity Code: **ELRNG01863**

#### Fundamentals of Generative AI [101-Basics]

This AI course is designed to equip learners with:

- The foundational knowledge and skills required to harness the power of Generative AI.
- The ability to identify opportunities for innovation and implementation of AI within their organizations.
- The skills to drive organizations toward a future of enhanced creativity and competitive advantage using AI techniques.

## Schedule – Stage 2: Day 27 to 60

Day 27 to 31 will be focusing on Selenium configuration and WebDriver Basics.

Udemy learnings are recommended in the Platform to understand the fundamental concepts.

Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

#### Integrated Project Deliverable: Agile and Functional Testing handson

As the selenium learning starts, the project details(By trainer) will be given for the learners so that you can parallel keep doing the project activities along with the rest of the learnings.

Please proceed with the automation implementation for the functional requirements provided in the previous milestone ("Trainer Given Handson").

Instructions: As you progress through this milestone's learning objectives (Selenium Architecture, Environment Setup, WebDriver, Locators, and Customized XPATH), incrementally integrate these newly acquired skills into your existing automation project.

**Objective:** The goal is to build a foundational, executable set of automated test scripts that validate the functionality defined in the agile documents you previously prepared.

The given Project are expected to be completed on or before the end week of the program.

Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

### **Business Aligned Project (Mini Project/ Hackathon)**

As the selenium learning starts, the project details (Mini project and the Hackathon) will be given for the learners so that they can parallel keep doing the project activities along with the rest of the learnings.

- Mini project in selenium is a great way to deepen your understanding, showcase your skills, and prepare for real-world automation tasks.
- It allows to apply theoretical knowledge of Selenium and web automation in a practical, real-world scenario. Working on a project helps you gain hands-on experience with Selenium WebDriver, locators, and other essential components.
- Building a mini project helps you understand the design and implementation of automation frameworks, such as Page Object Model (POM) and Data-Driven Testing.
- You can integrate Selenium with other tools like TestNG for test management and reporting.

### **Mini Project & Hackathon – Objective**

- CI/CD Integration: Integrating your Selenium tests with CI/CD tools like Jenkins helps automate the testing process as part of the build pipeline.
- Automated Testing: Ensures that your application is tested automatically with every code change, improving software quality.
- Reporting and Logging: Generate detailed test reports that provide insights into test execution, pass/fail status, and error details.
- Logging: Implement logging to capture detailed information about test execution, which is useful for debugging and analysis.
- Reusable Code: Writing reusable and maintainable code is a key aspect of any project, and a mini project helps you practice this.
- Collaboration: It encourages collaboration and teamwork, as you can divide tasks among team members and integrate their work using Selenium.
- Exception Handling: You'll learn to handle exceptions and errors that occur during test execution, making your scripts more robust.
- Debugging Skills: Debugging skills are enhanced as you troubleshoot issues in your automation scripts etc.

The given Mini Project and the Hackathon are expected to be completed on or before the end week of the program.

Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

**Note:**

Mini Project is an individual team member activity. Project Requirement is available in the platform. Candidates must complete the requirement and submit the solution for evaluations by BU SME.

Hackathon is an group activity. Project requirement is available in the platform. Candidates must complete the requirement and submit the solution for evaluations by BU SME.

| Selenium Automation for Realtime Project |                                    |                                                        |                                    |                                                                   |
|------------------------------------------|------------------------------------|--------------------------------------------------------|------------------------------------|-------------------------------------------------------------------|
| Section #                                | Module / Topic                     | Detailed Lecture / Sub-Topic                           | Concept/Tool                       | Notes                                                             |
| 1                                        | Setting the Project Foundation     | Automation Project Setup and Environment Configuration | Java, Maven/Gradle                 | Dependencies and Build Automation.                                |
|                                          |                                    | Introduction to Test Design Patterns                   | Page Object Model (POM)            | Structure, benefits, and implementation.                          |
|                                          |                                    | Setting up Logging and Reporting Tools                 | Log4j, Extent Reports              | Generating detailed, professional reports.                        |
| 2                                        | Advanced WebDriver Concepts        | Synchronization Techniques                             | Explicit & Fluent Waits            | Handling dynamic elements and improving stability.                |
|                                          |                                    | Handling Complex Web Elements                          | Action Class, JavaScript Executor  | Mouse and Keyboard interactions, scrolling.                       |
|                                          |                                    | Handling Popups, Windows, and Frames                   | WebDriver                          | Switching contexts and managing alerts.                           |
|                                          |                                    | Capturing and Analyzing Failure Points                 | Taking and Organizing Screenshots  | Capturing evidence upon test failure.                             |
| 3                                        | Designing the Automation Framework | Hybrid Framework Design                                | POM + Data-Driven + Keyword-Driven | Combining best practices into one framework.                      |
|                                          |                                    | Data Management for Live Projects                      | Data-Driven Testing                | Reading test data from Excel, JSON, or databases.                 |
|                                          |                                    | Implementing Robust Exception Handling                 | Java                               | Writing code that handles unexpected errors gracefully.           |
|                                          |                                    | Writing Reusable Utility Methods                       | Custom Libraries                   | Creating helper functions for common tasks (e.g., file handling). |
| 4                                        | Test Execution & Reporting         | Parallel Test Execution                                | TestNG/Selenium Grid               | Running multiple tests simultaneously to save time.               |
|                                          |                                    | Grouping and Prioritizing Tests                        | TestNG Annotations                 | Running Smoke, Sanity, or Regression Suites.                      |
|                                          |                                    | Finalizing and Interpreting Execution Reports          | Extent/Allure Reports              | Analyzing results for pass/fail metrics.                          |
| 5                                        | Real-Time DevOps Integration       | Integrating Framework with Version Control             | GIT/GitHub                         | Pushing and managing the project code.                            |
|                                          |                                    | Scheduling and Running Tests in CI                     | Jenkins                            | Setting up jobs for automated nightly/daily runs.                 |

## Day 27

### Automation Concepts, configuration, WebDriver Basics

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name                        | Topic                                                            |
|-----------|-------------------------------------|------------------------------------------------------------------|
| Chapter 1 | <b>Selenium Basics and overview</b> | Introduction to Browser Automation                               |
|           |                                     | What is Selenium and Overview                                    |
|           |                                     | History and evolution of Selenium                                |
|           |                                     | Selenium Components (WebDriver vs Selenium IDE vs Selenium Grid) |
|           |                                     | Supported programming languages (Java, Python, C#, etc.)         |
|           |                                     | Cross-browser and cross-platform compatibility.                  |
|           |                                     | Advantages and limitations of Selenium                           |

## Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Refer to and learn the same topics covered by the trainer on Udemy.
2. Practice the Tekstac Platform Hands-on.

This ensures a thorough understanding and application of the Selenium chapter topics.

## Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find Element by Id
- Find Element by Name

Learn the basics of WebDriver Basics



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Section 3: Selenium WebDriver

Note: You are requested to complete below Integrated Project Deliverable on or before your day 31

## Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

### Mandatory Hands-on

- Trainer-Guided Integrated Project: Automation Feasibility Report & Strategy
- Trainer-Guided Integrated Project: Configured Automation Environment & Project Structure
- Trainer-Guided Integrated Project: Automated Test Scripts (Individual Functional Scenarios)
- Trainer-Guided Integrated Project: Test Execution Reports and Status
- Trainer-Guided Integrated Project: Code Review and Feedback Session Artifacts

## Day 28

### Automation Concepts, configuration, WebDriver Basics

#### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name                           | Topic                                                                                                                                                                      |
|-----------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chapter 2 | <b>Selenium Architeture</b>            | What is Selenium Webdriver: 1) WebDriver is one of the component in selenium. 2) WebDriver is a java interface. 3) WebDriver is an API( Application Programming interface) |
|           |                                        | Understanding the WebDriver Architecture / API methods. WebDriver, RemoteWebDriver, ChromeDriver , FirefoxDriver, EdgeDriver etc..                                         |
| Chapter 3 | <b>Setting up Selenium environment</b> | Installing Java Development Kit (JDK) and Eclipse IDE                                                                                                                      |
|           |                                        | Downloading Selenium WebDriver                                                                                                                                             |
|           |                                        | Configuring Selenium WebDriver with Eclipse                                                                                                                                |

#### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning)hands-on exercises provided in the local environment.
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### **BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]**

- Chapter 2\_Handson
- Chapter 2\_PPT Learning Ref.Content
- Chapter 2\_Video Learning Ref.Content
- Chapter 3\_Handson
- Chapter 3\_PPT Learning Ref.Content
- Chapter 3\_Video Learning Ref.Content



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Section 3: Selenium WebDriver

Day 29

## Automation Concepts, configuration, WebDriver Basics

### Continuous Learning: Technical Enablement

| Chapter   | Chapter Name          | Topic                                                 |
|-----------|-----------------------|-------------------------------------------------------|
| Chapter 4 | Selenium<br>WebDriver | Locating GUI Elements in Application Under Test (AUT) |
|           |                       | How to identify the locators in web page              |
|           |                       | First Selenium Webdriver Script                       |
|           |                       | Launching Browsers in Selenium                        |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment Submit the solutions for evaluation.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### Continuous Learning: Technical Hands-on through Tekstac

#### Mandatory Hands-on

- Find options in dropdown
- Find Radio Element

## Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### **BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]**

- Chapter 4\_Handson
- Chapter 4\_PPT Learning Ref.Content
- Chapter 4\_Video Learning Ref.Content



#### [Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Section 3: Selenium WebDriver

Day 30

## Automation Concepts, configuration, WebDriver Basics

### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name                  | Topic                                                                                                      |
|-----------|-------------------------------|------------------------------------------------------------------------------------------------------------|
| Chapter 5 | Locating various Web elements | Find Element by Id,Name,Link Text,Partial Link Text,Class Name,CSS ( Tag , ID , Class , Attribute , etc..) |
|           |                               | Handling Shadow DOM in Selenium                                                                            |

## Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

## Continuous Learning: Technical Hands-on through Tekstac

### Mandatory Hands-on

- Find Element by Tag Name
- Handling Check Box
- Find Element by Class Name
- LinkText and PartialLinkText

### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 5\_Handson
- Chapter 5\_PPT Learning Ref.Content
- Chapter 5\_Video Learning Ref.Content[Part-1]
- Chapter 5\_Video Learning Ref.Content[Part-2]
- Chapter 5\_Video Learning Ref.Content[Part-3]



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Section 3: Selenium WebDriver

### Additional Learning:

### Technical Quizzes:

Quiz 1 - Automation Concepts, Selenium configuration, WebDriver Basics

Day 31

## Automation Concepts, configuration, WebDriver Basics

### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name                  | Topic                                                                                        |
|-----------|-------------------------------|----------------------------------------------------------------------------------------------|
| Chapter 5 | Locating various Web elements | Relative Locators and Chained Locators in Selenium                                           |
|           |                               | Locating By Xpath (Single Slash, Double Slash , contains(), starts_with() , text() , Last()) |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Practice the Assessment.

5. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Learn the basics of WebDriver Basics



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25.

Section 3: Selenium WebDriver

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Selenium-Web Driver** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

### Code Challenge

Assess-Type-1: Code Challenge - Automation Concepts, Selenium configuration, WebDriver Basics

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

Day 32

## Selenium Automation Techniques, Dynamic XPath

Click Here to View the [BU Mandatory Technical Learning and Hands-on](#) Document.

### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name                  | Topic                                                                                                            |
|-----------|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| Chapter 5 | Locating various Web elements | Find Element by Id, Name, Link Text, Partial Link Text, Class Name, CSS ( Tag , ID , Class , Attribute , etc.. ) |
|           |                               | Handling Shadow DOM in Selenium                                                                                  |
|           |                               | Relative Locators and Chained Locators in Selenium                                                               |
|           |                               | Locating By XPath (Single Slash, Double Slash , contains(), starts_with() , text() , Last())                     |
|           |                               | Find options in dropdown. Handle dropdown without using Select Class                                             |
|           |                               | Find Radio Element                                                                                               |
|           |                               | Find Element by Tag Name                                                                                         |
| Chapter 6 | Desired capabilities          | Desired Capabilities in Selenium WebDriver                                                                       |

## Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

## Continuous Learning: Technical Hands-on through Tekstac

### Mandatory Hands-on

- CSSLocator
- AbsoluteXpathLocator
- xPathAncestor
- Working with Alerts
- Work with Javascript Excecutor

**Note: You are requested to complete below Integrated Project Deliverable on or before your day 36**

### Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

#### Mandatory Hands-on

- Trainer-Guided Integrated Project: Enhanced Automation Test Scripts
- Trainer-Guided Integrated Project: Comprehensive Error Handling and Reporting
- Trainer-Guided Integrated Project: Utility Class for Advanced Functions
- Trainer-Guided Integrated Project: Code Review & Refactoring Artifacts

### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 6\_Handson
- Chapter 6\_PPT Learning Ref.Content
- Chapter 6\_Video Learning Ref.Content

Learn the basics of Automation Techniques and Dynamic Xpath



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

## Project – Mini Project Case Study

**Overall Duration:** Should start your mini project parallel with the Stage 2 Selenium Automation learnings.

The outcomes of doing **Mini Project** are:

- Enables learners to know on the environment setup
- Any web application is taken, and learner try to automate given scenario using Selenium APIs
- Exhibits learner skills on automation of real time applications for smaller requirement.

## Day 33

### Selenium Automation Techniques, Dynamic XPath

#### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name                                                          | Topic                                                            |
|-----------|-----------------------------------------------------------------------|------------------------------------------------------------------|
| Chapter 7 | selenium<br>webdriver<br>commands and<br>Web elements<br>interactions | Selenium WebDriver- Commands                                     |
|           |                                                                       | Text fields and buttons                                          |
|           |                                                                       | Find Element VS FindElements in Selenium WebDriver - check order |
|           |                                                                       | Handling Canvas Elements in Selenium                             |
|           |                                                                       | Using JavaScript Executor in Selenium                            |

#### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

#### Continuous Learning: Technical Hands-on through Tekstac

##### Mandatory Hands-on

- Handling RegEx Selenium
- Drag and Drop with Slider and Actions and wait
- Selenium – ScreenShot
- Selenium - DatePicker(Actions Class)
- FindElements

**BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]**

- Chapter 7\_Handson
- Chapter 7\_PPT Learning Ref.Content
- Chapter 7\_Video Learning Ref.Content

Learn the basics of Automation Techniques and Dynamic Xpath



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

## Project– Mini Project Case Study

- Integrate your mini project

## Stage : GenAI Code Assist (GitHub Copilot)

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **GenAI Foundation**

This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.

- Fundamentals of Generative AI [101-Basics]
- Generative AI Quick Assessment for Elearning Quiz [101-Basics]
- Introduction to Responsible AI
- Responsible AI-Deep Dive [101- BASICS]

- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.

- Prompt Engineering Foundation - PY0009562

- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.

Introduction to GitHub Copilot



[GitHub Copilot for Beginners: AI Coding Crash Course](#)

Go through entire course. Implement the examples along with the author.



## Usecase and Handbook & Comprehensive GenAI Hands-On for Automation

**Developers:** These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional, high-quality automation code, making the learning directly applicable to their job roles.

### Day 34

## Selenium Automation Techniques, Dynamic XPath

### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name      | Topic                              |
|-----------|-------------------|------------------------------------|
| Chapter 8 | WebDriver Actions | Mouse and keyboard actions         |
|           |                   | Drag and drop                      |
|           |                   | Handling frames and windows        |
|           |                   | mouse hover (tool tip) in Selenium |
|           |                   | Handling tabs & cookies            |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### Continuous Learning: Technical Hands-on through Tekstac

#### Mandatory Hands-on

- Handling ToolTip
- Working with different XPath conditions
- PurchaseApp iframe
- Multiple Window
- Selenium - Implicit Wait
- Selenium - Fluent Wait

## Continuous Learning: Technical Hands-on

### Best Practices- Real Time Technical Hands-on

- Desired Capabilities
- Shadow\_DOM\_Scenario1
- Shadow\_DOM\_Scenario2
- Relative Locators and Chained Locators in Selenium

### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 8\_Handson
- Chapter 8\_PPT Learning Ref.Content
- Chapter 8\_Video Learning Ref.Content

Learn the basics of Automation Techniques and Dynamic Xpath



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

### Project – Mini Project Case Study

- Integrate the mini project.

### Stage : GenAI Code Assist (GitHub Copilot)

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **GenAI Foundation**

This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.

- Fundamentals of Generative AI [101-Basics]
- Generative AI Quick Assessment for Elearning Quiz [101-Basics]
- Introduction to Responsible AI
- Responsible AI-Deep Dive [101- BASICS]

- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.

- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.

Introduction to GitHub Copilot



[GitHub Copilot for Beginners: AI Coding Crash Course](#)

Go through entire course. Implement the examples along with the author.



[GitHub Copilot - The Complete Guide – 2025](#)

Go through entire course. Implement the examples along with the author.

### Usecase and Handbook & Comprehensive GenAI Hands-On for Automation

**Developers:** These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional, high-quality automation code, making the learning directly applicable to their job roles.

## Day 35

### Selenium Automation Techniques, Dynamic XPath

#### Continuous Learning: ILT Technical Enablement

| Chapter   | Chapter Name              | Topic                         |
|-----------|---------------------------|-------------------------------|
| Chapter 9 | Multiple windows & frames | Switching between windows     |
|           |                           | Switching to frames & iFrames |
|           | WebDriver Waits           | Implicit waits                |
|           |                           | Explicit waits                |
|           |                           | Fluent waits                  |

| Chapter    | Chapter Name                          | Topic                                                                                                 |
|------------|---------------------------------------|-------------------------------------------------------------------------------------------------------|
| Chapter 10 | <b>Selenium Automation Techniques</b> | Checkbox Handling and Form Submission                                                                 |
|            |                                       | Locating Web Element Attributes using CSS Selector                                                    |
|            |                                       | Locating Web Element Attributes using Absolute XPath Locator                                          |
|            |                                       | Identifying Ancestors of Web Elements                                                                 |
|            |                                       | Working with alerts                                                                                   |
|            |                                       | Working with Java script executor                                                                     |
|            |                                       | Working with Regular expression                                                                       |
|            |                                       | Working with Drag and Drop and Slider                                                                 |
| Chapter 11 | <b>Advanced WebDriver techniques</b>  | Selenium - ScreenShot                                                                                 |
|            |                                       | Handling JavaScript/window Alerts                                                                     |
|            |                                       | Headless browsers and drivers in Selenium                                                             |
|            |                                       | Handling Dynamic Elements in Selenium                                                                 |
|            |                                       | 1) get methods , 2) conditional methods, 3) browser methods, 4) navigational methods, 5) wait methods |
|            |                                       | How to Find All/Broken links using Selenium Webdriver                                                 |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Take Up the Assessment.
5. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### **BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]**

- Chapter 9\_Handson
- Chapter 9\_PPT Learning Ref.Content
- Chapter 9\_Video Learning Ref.Content
- Chapter 10\_Handson
- Chapter 10\_PPT Learning Ref.Content
- Chapter 10\_Video Learning Ref.Content
- Chapter 11\_Handson
- Chapter 11\_PPT Learning Ref.Content
- Chapter 11\_Video Learning Ref.Content

### Continuous Learning: Technical Hands-on through Tekstac

#### **Best Practices- Real Time Technical Hands-on**

- Relative Locators and Chained Locators in Selenium\_Scenario 1
- Relative Locators and Chained Locators in Selenium\_Scenario 2
- Relative Locators and Chained Locators in Selenium\_Scenario 3

- Relative Locators and Chained Locators in Selenium\_Scenario 4



### [Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

#### **Additional Learning:**

#### **Technical Quizzes:**

Quiz 1 - Selenium Automation Techniques, Dynamic XPath

#### **Project– Mini Project Case Study**

- Integrate your mini project

Day 36

### **Selenium Automation Techniques, Dynamic XPath**

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Selenium – XPath** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

#### **Code Challenge**

Assess-Type-1: Code Challenge - Selenium Automation Techniques, Dynamic XPath

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

#### **Project – Mini Project Case Study**

- Integrate your mini project

You will be focusing on Selenium WebDriver with POM and ApachePOI

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

## Selenium Webdriver With POM and ApachePOI

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

### Continuous Learning: ILT Technical Enablement

| Chapter    | Chapter Name          | Topic                                               |
|------------|-----------------------|-----------------------------------------------------|
| Chapter 12 | Calendar using action | Selenium - DatePicker(Actions Class)                |
|            | Apache POI            | Apache POI - Read Operation : Excel Read Operation  |
|            |                       | Apache POI - Write Operation: Excel Write Operation |
|            |                       | Applying POI POM- Flight reservation                |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### Continuous Learning: Technical Hands-on through Tekstac

#### Mandatory Hands-on

- Excel Read Operation
- Excel Write Operation
- Applying POI - Flight reservation

#### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 12\_Handson
- Chapter 12\_PPT Learning Ref.Content
- Chapter 12\_Video Learning Ref.Content

Learn the basics of WebDriver with POM and ApachePOI



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 4 in this Udemy course and complete the corresponding learnings.

Section 4: Apache POI - Data Driven Testing using MS Excel

Note: You are requested to complete below Integrated Project Deliverable on or before your day 41

### **Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation**

#### **Mandatory Hands-on**

- Trainer-Guided Integrated Project: Refactored Automation Framework (Page Object Model Implementation)
- Trainer-Guided Integrated Project: Data Management Strategy & Implementation (Apache POI)
- Trainer-Guided Integrated Project: Execution Confirmation and Custom Logging Artifacts
- Trainer-Guided Integrated Project: Updated Agile Artifacts & Traceability (Living Documentation)

### **Project – Mini Project Case Study**

- Integrate the mini project.

## **Day 38**

### **Selenium Webdriver With POM and ApachePOI**

Learn the basics of WebDriver with POM and ApachePOI



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 4 in this Udemy course and complete the corresponding learnings.

Section 4: Apache POI - Data Driven Testing using MS Excel.

### **Continuous Learning: Technical Hands-on through Tekstac**

#### **Mandatory Hands-on**

- POM with Page Factory Model

## Continuous Learning: Technical Hands-on

### Best Practices- Real Time Technical Hands-on

- Mouse\_Keyboard\_Actions\_Cookie\_Scenario 1
- Mouse\_Keyboard\_Actions\_Cookie\_Scenario 2
- Mouse\_Keyboard\_Actions\_Cookie\_Scenario 3
- Mouse\_Keyboard\_Actions\_Cookie\_Scenario 4
- Mouse\_Keyboard\_Actions\_Cookie\_Scenario 5
- Mouse\_Keyboard\_Actions\_Cookie\_Scenario 6
- Mouse\_Keyboard\_Actions\_Cookie\_Scenario 7

### Project – Mini Project Case Study

- Integrate the mini project.

Day 39

## Selenium Webdriver With POM and ApachePOI

### Continuous Learning: Technical Enablement

Learn the basics of WebDriver with POM and ApachePOI



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 4 in this Udemy course and complete the corresponding learnings.

Section 4: Apache POI - Data Driven Testing using MS Excel

Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Selenium – Apache POI** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

### Code Challenge

Assess-Type-1:Code Challenge - Selenium Webdriver with POM and ApachePOI

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

### Project – Mini Project Case Study

- Integrate the mini project.

## Automation Testing - Selenium with TestNG

You will be focusing on Selenium with TestNG

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

### Continuous Learning: ILT Technical Enablement

| Chapter    | Chapter Name    | Topic                                  |
|------------|-----------------|----------------------------------------|
| Chapter 13 | TestNG Concepts | TestNG Assertions                      |
|            |                 | Introduction to TestNG                 |
|            |                 | Creating TestNG test cases             |
|            |                 | TestNG annotations and assertions      |
|            |                 | Data-Driven Testing in TestNG          |
|            |                 | Running Tests in Parallel using TestNG |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- TestNG Assertions
- TestNG with two test classes
- Contact Book- POM structure
- TestNG Annotation With Priority
- TestNG Dependency

### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 13\_Handson
- Chapter 13\_PPT Learning Ref.Content
- Chapter 13\_Video Learning Ref.Content

Learn the basics of Selenium with TestNG



Refer section 5,6,7 in this Udemy course and complete the corresponding learnings.

Section 5: TestNG Framework for Selenium

Section 6: Page Object Model

Section 7: Hybrid Driven Framework Development with Project (eCommerce Application)

**Note: You are requested to complete below Integrated Project Deliverable on or before your day 43**

### **Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation**

#### **Mandatory Hands-on**

- Trainer-Guided Integrated Project: Integrated & Structured Test Suite using TestNG Framework
  - ✓ You should have automated more than one sprints
  - ✓ Ensure you have in-sprint automation concepts introduced as part of project
  - ✓ Ensure you have created Regression suite and smoke suite automation
  - ✓ Ensure your regression suite consists of minimum 25 test cases and test scripts
  - ✓ Ensure your smoke suite covers end to end business functionality
- Trainer-Guided Integrated Project: Robust Test Data Management via TestNG DataProviders
- Trainer-Guided Integrated Project: Comprehensive Execution Control via testng.xml
- Trainer-Guided Integrated Project: Advanced Reporting with Extent Reports & Listeners
- Trainer-Guided Integrated Project: Implementation of Assertions (Hard and Soft)

## Automation Testing - Selenium with TestNG

### Continuous Learning: ILT Technical Enablement

| Chapter    | Chapter Name    | Topic                                                        |
|------------|-----------------|--------------------------------------------------------------|
| Chapter 13 | TestNG Concepts | Multi-browser testing using TestNG*                          |
|            |                 | Rerun failed tests*                                          |
|            |                 | Set Test priority in TestNG                                  |
|            |                 | Dependency in TestNG                                         |
|            |                 | Soft /Hard Assertion in TestNG                               |
|            |                 | TestNG Report Generation in Selenium WebDriver*              |
|            |                 | Customize, PDF & Email TestNG Reports in Selenium WebDriver* |
|            |                 |                                                              |
|            | TestNG Class    | TestNG with two test classes                                 |
|            | Data driven     | Data driven from XML + POM : Contact Book- POM structure     |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### Continuous Learning: Technical Hands-on through Tekstac

#### Mandatory Hands-on

- Cargo Shipping
- Shopify Registration
- XMLParsing -1
- XMLParsing -2
- XMLParsing -3



## [Learn API Technical Writing: JSON and XML for Writers](#)

Refer sections 1, 2, 3 and 4 in this Udemy course and complete the corresponding learnings.

Section 1: Introduction

Section 2: JSON

Section 3: XML

Section 4: Final Words

Rehearse the below Lend a hand enablement given in platform

- XMLParser\_Enablement

Followed by refer “Lend-a-Hand” code template with solution

- XML Parsing

Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Selenium – TestNG** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

**Code Challenge** - Assess-Type-1:Code Challenge - Selenium with TestNG

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

## **Project – Mini Project Case Study**

- Integrate the mini project.

Day 42 and 43

## **Automation Testing - Selenium with TestNG**

### **Continuous Learning: Technical Enablement**

Learn the basics of Selenium with Datasource JSON Parsing (With XML & JSON)

Rehearse the below Lend a hand enablement given in platform

- JSONParsing\_Enablement

Followed by refer “Lend-a-Hand” code template with solution

- JSON Parsing
- Json Solution Explanation

## Continuous Learning: Technical Hands-on through Tekstac

### Mandatory Hands-on

- JSONParsing – 1
- JSONParsing – 2
- JSONParsing – 3

**NOTE: On Day 42**, Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Selenium** skill appear for the **MOCK ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

### Integrated Capability Test (ICT) - SELENIUM MOCK

Assessment Duration: 2 hrs.

Assessment Attempts: 1

No.of Questions appear in each attempt : 1

## Integrated Capability Test (ICT) - SELENIUM

### SELENIUM Assess-Type-2: Integrated Capability Test (ICT)

**NOTE: On Day 43**, Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Selenium** skill appear for the **Selenium ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assessment Duration: 4 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

### Project – Mini Project Case Study

- Integrate the mini project.

**Deliver your project and get the review comments from SME.**

## Day 44 & 45

### Evaluation:

#### Interim Project Evaluation + Technical Evaluation

The overall Interim Technical evaluation will cover topics specified up to the Interim Evaluation in the curriculum and will be conducted via a video interview on the GenC Learn Platform.

There are no Additional Attempts for the Interim Evaluation. RAG Achieving in the first attempt is considered as final RAG for Interim Evaluation.

## Automation Framework

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

### Continuous Learning: ILT Technical Enablement

| Chapter    | Chapter Name                      | Topic                                                                       |
|------------|-----------------------------------|-----------------------------------------------------------------------------|
| Chapter 14 | <b>Test Automation Frameworks</b> | Introduction to Automation Frameworks                                       |
|            |                                   | Data-Driven Testing in Selenium                                             |
|            |                                   | Keyword-Driven Testing Framework in Selenium                                |
|            |                                   | Page Factory                                                                |
|            |                                   | Building a Modular Testing Framework in Selenium                            |
|            |                                   | Hybrid Framework                                                            |
|            | <b>Design Patterns (POM)</b>      | Understanding Page Object Model (POM) in Selenium                           |
|            |                                   | Creating Page Classes in Selenium                                           |
|            |                                   | Implementing Page Factory in Selenium                                       |
|            |                                   | Design Patterns in Automation using Selenium (Singleton, Factory etc.)      |
| Chapter 15 | <b>Test Data Handling</b>         | Reading and Writing Data to Files (e.g., Excel, CSV , XML, JSON) using Java |
|            |                                   | Test Data Parameterization in Selenium                                      |
| Chapter 16 | <b>Test Report and Test Log</b>   | Generating Detailed Test Reports (Extent Reports, Allure Reports)           |
|            |                                   | Screenshot Capture Strategies in Selenium                                   |
|            |                                   | Custom Logging with Log4j in Selenium                                       |

### Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

### **BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]**

- Chapter 14\_Handson
- Chapter 14\_PPT Learning Ref.Content
- Chapter 14\_Video Learning Ref.Content[Part-1]
- Chapter 14\_Video Learning Ref.Content[Part-2]
- Chapter 14\_Video Learning Ref.Content[Part-3]

- Chapter 15\_Handson
- Chapter 15\_PPT Learning Ref.Content
- Chapter 15\_Video Learning Ref.Content
- Chapter 16\_Handson
- Chapter 16\_PPT Learning Ref.Content
- Chapter 16\_Video Learning Ref.Content[Part-1]
- Chapter 16\_Video Learning Ref.Content[Part-2]
- Chapter 16\_Video Learning Ref.Content[Part-3]

**Note: You are requested to complete below Integrated Project Deliverable on or before your day 54**

### **Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation**

- Trainer-Guided Integrated Project: Scalable Execution Infrastructure (Selenium Grid & Docker)
- Trainer-Guided Integrated Project: Behavior-Driven Development (BDD) Framework Integration
- Trainer-Guided Integrated Project: Continuous Integration & Delivery (CI/CD) Pipeline Integration

## **Day 47 to Day 54**

### **Devops**

You will be focusing on Devops (Maven, Git, Jenkins, Grid).

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

**Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.**

### **Table of Contents:**

| Topic #                           | Topic Name | Topic Objective                         | References                                  |
|-----------------------------------|------------|-----------------------------------------|---------------------------------------------|
| Behavior-Driven Development (BDD) | Cucumber   | Introduction to BDD and Gherkin Syntax  | Writing feature files and step definitions. |
|                                   | Cucumber   | Integrating BDD with Selenium Framework | Running scenarios and hooks.                |
| Build Automation & Dependencies   | Maven      | Introduction to Maven                   | Understanding Project Object Model (POM).   |

|                                 |               |                                                |                                                                  |
|---------------------------------|---------------|------------------------------------------------|------------------------------------------------------------------|
|                                 | Maven         | Managing Dependencies and Plugins              | Setting up Selenium and other libraries.                         |
|                                 | Maven         | Executing Tests using Maven Goals              | Running tests from the command line.                             |
| Version Control System          | GIT           | Introduction to GIT and GitHub                 | Basic commands (clone, add, commit, push, pull).                 |
|                                 | GIT           | Branching and Merging Strategies               | Collaborative development practices.                             |
|                                 | GitHub        | Integrating Test Code with Remote Repository   | Pushing the automation framework to the cloud.                   |
| Continuous Integration/Delivery | Jenkins       | Setting up Jenkins CI Server                   | Installation and configuration.                                  |
|                                 | Jenkins       | Creating a Jenkins Pipeline                    | Integrating GIT and Maven projects.                              |
|                                 | Jenkins       | Scheduling and Triggering Automation Runs      | Generating test reports in CI environment.                       |
| Distributed Test Execution      | Selenium GRID | Understanding Selenium Grid Architecture       | Hub and Node concepts.                                           |
|                                 | Selenium GRID | Setting up and Configuring the Grid            | Running tests in parallel on multiple machines/browsers.         |
| Containerization                | Docker        | Introduction to Docker                         | Understanding images and containers.                             |
|                                 | Docker        | Running Selenium Tests using Docker Containers | Setting up Dockerized Selenium Grid for consistent environments. |
|                                 | Docker        | Creating a Custom Automation Image             | Packaging the complete framework.                                |

## Day 47, Day 48 & Day 49

### Devops (GIT, Maven, Grid, Jenkins)



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections 8 in this Udemy course and complete the corresponding learnings.

Section 8: CI - Maven, Git, GitHub & Jenkins

## Continuous Learning: ILT Technical Enablement

| Chapter    | Chapter Name          | Topic                                                                         |
|------------|-----------------------|-------------------------------------------------------------------------------|
| Chapter 17 | Selenium Grid         | Introduction to Selenium Grid                                                 |
|            |                       | Setting up Selenium Grid                                                      |
|            |                       | Running tests on Selenium Grid                                                |
|            |                       | Achieving Parallel Execution in the Cloud                                     |
|            |                       | Setting up selenium grid in a docker/K8S container                            |
|            | Distributed Execution | Running Tests on Cloud-Based Selenium Grids (e.g., Sauce Labs, Browser Stack) |
|            |                       | Achieving Parallel Execution in the Cloud                                     |
|            | Maven                 | Introduction to Maven                                                         |
|            |                       | Creating and Managing Maven Projects                                          |
|            |                       | Maven Build Lifecycle (e.g., clean, compile, test, package)                   |
|            | Git Version Control   | Introduction to Version Control                                               |
|            |                       | Git Fundamentals (Commits, Branches, Merges)                                  |
|            |                       | Git Workflow (Pull Requests)                                                  |
|            |                       | Managing Code Repositories on GitHub or GitLab                                |
|            | Jenkins CI Systems    | Introduction to CI/CD and DevOps                                              |
|            |                       | Jenkins Overview and Jobs Configuration                                       |
|            |                       | Integrating Selenium Tests into Jenkins Pipelines                             |
|            |                       | Groovy Script Introduction and Syntax                                         |

## Continuous Learning: Technical Hands-on through Tekstac

### Mandatory Hands-on

- Git Config
- Clone Repo
- Add, Commit and Push
- Pull And Merge
- Merge - Resolve Conflict
- Git Tags
- Build project based on another project
- Build Maven Project (Using Freestyle Project)

### Additional Hands-on

- GIT installation & commands implementation
- Install Jenkins and Creation of new job

### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 17\_Handson
- Chapter 17\_PPT Learning Ref.Content
- Chapter 17\_Video Learning Ref.Content[Part-1]
- Chapter 17\_Video Learning Ref.Content[Part-2]
- Chapter 17\_Video Learning Ref.Content[Part-3]

- Chapter 17\_Video Learning Ref.Content[Part-4]
- Chapter 17\_Video Learning Ref.Content[Part-5]

## Day 50 & Day 51



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections 8 in this Udemy course and complete the corresponding learnings.

Section 8: CI - Maven, Git, GitHub & Jenkins

### Continuous Learning: Technical Hands-on through Tekstac

#### Mandatory Hands-on

- Pipeline – compile and test
- Build Web Application
- Compiling Executing Java using POM

### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 18\_Handson
- Chapter 18\_PPT Learning Ref.Content
- Chapter 19\_Handson
- Chapter 19\_PPT Learning Ref.Content

#### Additional Hands-on

- Mavenization using Eclipse M2E plugin Options and CLI command

### Project – Hackathon

Integrate the Hackathon project and work on daily basis

## Day 52, Day 53 and Day 54



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections 8 in this Udemy course and complete the corresponding learnings.

Section 8: CI - Maven, Git, GitHub & Jenkins

## Continuous Learning: Technical Hands-on

### Mandatory Hands-on

- Junit With Maven
- Maven Directory Structure
- Maven Shade Plugin
- Maven App to access External Service
- Capital Service
- Setup Grid and parallel execution

### BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 20\_Handson
- Chapter 21\_Handson
- Chapter 21\_PPT Learning Ref.Content
- Chapter 22\_Handson
- Chapter 23\_Handson
- Chapter 24\_Handson

### Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 55 to Day 56

### BDD Cucumber

#### Continuous Learning: ILT Technical Enablement



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 9 in this Udemy course and complete the corresponding learnings.

Section 9: Cucumber BDD Framework

## Continuous Learning: Technical Hands-on

### Mandatory Hands-on

- Address Book
- Customer Registration

### Project – Hackathon

## Day 57 to Day 61

Day 57 to Day 61 we will learn API Automation

### API Automation

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

**Note: you are expected to complete below project deliverables on or before day 61**

**Continuation of Integrated Project Deliverable: Agile and Functional Testing hands-on through Automation**

- Trainer-Guided Integrated Project: The "API Smoke Suite" Postman Collection
- Trainer-Guided Integrated Project: The "Rest Assured" Automation Module (Framework Integration)
- Trainer-Guided Integrated Project: Comprehensive Automation Strategy & Traceability
- Trainer-Guided Integrated Project: Consolidated Execution Reports

## Day 57

### POSTMAN

**Continuous Learning: ILT Technical Enablement**

Refer Section 2 and Section 3 in the below Udemy course to understand the concepts



[Webservices API Testing with Rest Assured API & POSTMAN 2024](#)

**Continuous Learning: Technical Hands-on**

**Mandatory Hands-on**

- User Service 1 - Test Using Postman
- User Service 2 - Test Using Postman
- User Service 3 - Test Using Postman
- Cinema Hall Service 1 - Test Using Postman
- Find Winner SOAP Webservice - Test Using Postman

# API Automation using RESTASSURED

## Continuous Learning: Technical Enablement

Learn the API RESTASSURED Basic concepts



[REST Assured API Automation, Allure Reports - An Introduction](#)

## Chapter\_1\_API\_Fundamentals

### Mandatory Hands-on

- Verify the Success Status code 200 & 404 for the Bitcoin rate against USD and Non-USD currencies.
- Print Bitcoin rate against USD, GBP & EuR currency

## Chapter\_2\_API\_Fundamentals

### Mandatory Hands-on

#### Chapter\_2\_API using Postman – Part 1

- Download Postman
- Create workspace for API Request
- Create the Collections for API Request
- Sending the first API Request

#### Chapter\_2\_API using Postman – Part 2

- Get the response of GET API Request
- Get the response of POST API Request
- Get the response of PUT API Request

#### Chapter\_2\_API using Postman – Part 3

- Get the response of PATCH and DELETE API Request
- Create an environment in Postman
- Set the environment variables in Postman

#### Chapter\_2\_API using Postman – Part 4

- Get the values of the environment variables in Postman
- Get the ID of the given API in Postman
- Data Parameterization using CSV in Postman
- Test API Using Swagger UI

## Chapter\_3\_API\_Fundamentals

## Mandatory Hands-on

### Chapter\_3\_API in RestAssured – Part 1

- Verify the response of the GET API
- Validate the 200 status code in Rest Assured
- Validate the 301 status code in Rest Assured
- Validate the 404 status code in Rest Assured

### Chapter\_3\_API in RestAssured – Part 2

- Validate the 500 status code in Rest Assured
- Get the API response in JSON in Rest Assured
- Get the API response in XML in Rest Assured
- Get the API response with single query parameter in Rest Assured

### Chapter\_3\_API in RestAssured – Part 3

- Get the API response with double query parameter in Rest Assured
- Get the API response with path parameter in Rest Assured
- Get the API response with base URI in Rest Assured

## Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 58

## API Automation using RESTASSURED

### Continuous Learning: Technical Enablement

Learn the API RESTASSURED concepts



[REST Assured API Automation, Allure Reports - An Introduction](#)

### Chapter\_4\_API\_Fundamentals

#### Mandatory Hands-on

#### Chapter\_4\_RestAssured – Part 1

- Create workspace for API Request
- Create the Collections for API Request
- Sending the first API Request
- Get the response of GET API Request
- Get the response of POST API Request

## **Chapter\_4\_RestAssured – Part 2**

- Get the response of GET API Request
- Get the response of PUT API Request
- Get the response of PATCH API Request
- Get the response of DELETE API Request
- Create an environment in Postman

## **Chapter\_4\_RestAssured – Part 3**

- Set the environment variables in Postman
- Get the values of the environment variables in Postman
- Get the ID of the given API in Postman
- Data Parameterization using CSV in Postman

## **Chapter\_6\_API\_Fundamentals**

### **Mandatory Hands-on**

- Verify the Success Status code 200 for Add product details request
- Print the success response for Add product details request
- Update price of existing product details and verify the success status code 200
- Update price of existing product details and print the updated price from response
- Update only name of the existing product and verify the success code 200 displayed
- Delete the product which is not available in the server and verify the status code 404 and 200

## **Chapter\_7\_API\_Fundamentals**

### **Mandatory Hands-on**

- Verify the response status code with status code method in Rest Assured
- Verify the response status code with assert That in Rest Assured
- Verify the response status code with assertion with status Line method in Rest Assured
- Verify the response status code with assertion with Hamcrest matchers method in Rest Assured
- Handle the response using Getting a single header in Rest Assured
- Handle the response using Getting all headers in Rest Assured
- Validating a header in Rest Assured
- Validating multiple headers in Rest Assured
- Validating JSON schema using matchesJsonSchemaIn Classpath in Rest Assured

- Field-by-Field Validation in Rest Assured
- Structural Validation in Rest Assured
- Validate the inline schema in rest assured
- Validating response using Equality Check in Rest Assured
- Validating response using List Size Check in Rest Assured
- Validating response using Existence Check in Rest Assured

## Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 59

## API Automation using RESTASSURED

### Continuous Learning: Technical Enablement

Learn the API RESTASSURED concepts



[REST Assured API Automation, Allure Reports - An Introduction](#)

### Chapter\_7\_API\_Fundamentals

#### Mandatory Hands-on

- Validating response using Multiple Assertions in Rest Assured
- Validating response using Logical Operations in Rest Assured
- Validating response using Custom Parsers in Rest Assured
- Validating response for a List of Values in Rest Assured
- Validate cookies in Rest Assured
- Get the response of an API using single header in Rest Assured
- Get the response of an API using multiple headers in Rest Assured
- Get the response of an API by setting headers from a map in Rest Assured
- Utilize the redirect false command in Rest Assured
- Limit the number of redirects in Rest Assured
- Follow redirects but not use relaxed HTTPS validation on the redirected requests in Rest Assured

### Chapter\_8\_API\_Fundamentals

#### Mandatory Hands-on

- Ensure JSON schema data types are matching for the jsonplaceholder API request.

- Ensure JSON schema data types are not matched for the jsonplaceholder API request.
- Print number of universities in United States

### **BU Mandatory Handson**

- Print number of universities in India
- Print number of universities in Canada
- Print the university names in United States
- Print the university names in India
- Print the university names in Canada
- Store Uganda university details response into a json file
- Store Spain university details response into a text file
- Print second available university name in France
- Verify the United States Population is "318558162" in the year of 2016
- Verify the age is "34" for the name="meelad"

## **Chapter\_9\_API\_Fundamentals**

### **Mandatory Hands-on**

- Verify the success status code and response body from a API using Basic Authentication in Postman
- Verify the success status code and response body from an API using Basic Authentication in RestAssured
- Verify the failure status code and response body from an API using Basic Authentication in Postman

### **BU Mandatory Handson**

- Verify the failure status code and response body from an API using Basic Authentication in RestAssured
- Verify the success status code and response body from an API using Digest Authentication in Postman
- Verify the success status code and response body from an API using Digest Authentication in RestAssured
- Verify the failure status code and response body from an API using Digest Authentication in Postman
- Verify the failure status code and response body from an API using Digest Authentication in RestAssured
- Verify the success status code and response body of an API using API Key Authentication in Postman
- Verify the success status code and response body of an API using API Key Authentication in RestAssured Framework
- Verify the failure status code and response body of an API using API Key Authentication in Postman
- Verify the failure status code and response body of an API using API Key Authentication in RestAssured Framework
- Verify the success status code and response body of an API using Bearer token in Postman

- Verify the success status code and response body of an API using Bearer token in Rest Assured
- Verify the Failure status code and response body of an API using Bearer token in Postman
- Verify the failure status code and response body of an API using Bearer token in Rest Assured
- Verify the success status code and response body of an API using OAuth 2 token in Postman
- Verify if the response is successfully generated for an API using OAuth Authentication type
- Verify if the response is successfully generated for an API using OAuth Authentication type

## Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 60

## API Automation using RESTASSURED

### Continuous Learning: Technical Enablement

Learn the API RESTASSURED concepts



[REST Assured API Automation, Allure Reports - An Introduction](#)

### Chapter\_10\_API\_Fundamentals

#### Mandatory Handsons:

- Get number of universities in United States and print the response log
- Get number of universities in India and print response log by using ResponseLoggingFilter class
- Get the number of universities in Canada and write response log into text file

#### BU Mandatory Hands-on

- Get the university names in United States and print response log into ExtentReport
- Verify whether error status code 404 is displayed
- Verify whether except status code 200 is displayed
- Validate the response body for detailed error message. Ensure "404 Not Found" error response message is received

- Log the error details by using ErrorLoggingFilter class
- Handle the Generic exeception and print the same
- Handle the Assertion exeception and print the same.

## **Chapter\_11\_API\_Fundamentals**

### **Mandatory Hands-on**

- Print United States population details for the year 2022.Use Request Specification for Base URI,drilldowns & measures
- Print United States population details for the year 2020.Overwrite Request Specification for Base URI, drilldowns & measures
- Print queried Request Specification by using Query() method

## **Chapter\_12\_API\_Fundamentals**

### **Mandatory Hands-on**

- Print United States population details for the year 2022.Use Response Specification for status code & Content type as JSON
- Print United States population details for the year 2020.Overwrite Response Specification for Status code & ContentType
- Print United States population details for the year 2019. Use ResponseSpecBuilder for Status code & ContentType

## **Chapter\_13\_API\_Fundamentals**

### **Mandatory Hands-on**

- Create a serilization POJO class in Rest Assured framework
- Create a serilization class in Rest Assured framework
- Create a de-serilization POJO class in Rest Assured framework

#### **BU Mandatory Hands-on**

- Create a de-serilization class in Rest Assured framework

## **Chapter\_14\_API\_Fundamentals**

### **Mandatory Hands-on**

- Directly hardcode the json body data in the string variable, use it for post request and verify the success status code 200
- Create a separate class file, create a static string variable,store the json data,use it for post request and verify the success status code 200
- Send A JSON file as payload to the request and verify the success status code 200

#### **BU Mandatory Hands-on**

- Send A XML file as payload to the request and verify the success status code 200
- Send a request with the Content-Type of form-data parameters and verify the success status code 200

- Upload a png file into server and verify the success status code 200

## Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 61

## API Automation using RESTASSURED

### Continuous Learning: Technical Enablement

Learn the API RESTASSURED concepts



[REST Assured API Automation, Allure Reports - An Introduction](#)

### Chapter\_15\_API\_Fundamentals

#### Mandatory Hands-on

- Verify the success response of the upload API from postman
- Verify the failure response of the upload API from postman
- Verify the success response of the upload API using multipart from rest assured

#### BU Mandatory Hands-on

- Verify the success response of the download API from postman
- Verify the failure response of the upload API from postman 1
- Verify the success response of the upload API using multipart from rest assured 1

### Chapter\_16\_API\_Fundamentals

#### Mandatory Hands-on

- Verify the create user API and extract the ID value from Create API in Postman
- Verify if the response values from the API is getting stored in environment variables in postman
- Verify the create user API in RestAssured using faker

#### BU Mandatory Hands-on

- Verify the create user API in Rest Assured using dynamic data generation such as data from templates
- Verify the create user API in Rest Assured and pass the value of ID using ITest Context

- Verify the Get user API and extract the ID value from Create API in Postman
- Verify the GET user API in Rest Assured and get the value of ID using ITest Context
- Verify the Update user API and extract the ID value from Create API in Postman
- Verify the UPDATE user API in Rest Assured and get the value of ID using ITest Context
- Verify the Delete user API and extract the ID value from Create API in Postman
- Verify the UPDATE user API in Rest Assured and get the value of ID using ITest Context

## Chapter\_17\_API\_Fundamentals

### Mandatory Hands-on

1. Create a cucumber BDD Framework which should support API and Web automation.

## Chapter\_18\_API\_Fundamentals

### Mandatory Hands-on

- Working with GitHub and Git
- Integration of Rest assured API test script(Maven/TestNG) Based framework with jenkins Freestyle project
- Execute different methods parallelly using TestNG in Rest Assured

### BU Mandatory Hands-on

- Execute different suites parallelly using TestNG in Rest Assured
- Execute different classes parallelly using TestNG in Rest Assured
- Integrate and generate Extent reports in Rest Assured Framework

## Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 62

## Docker & Container

### Continuous Learning: Technical Enablement



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections 13 in this Udemy course and complete the corresponding learnings.

## GenAI Code Assist (GitHub Copilot)

### Comprehensive GenAI Hands-On for Automation Developers

- Trainer-Guided Integrated Project: Chat Endpoint Basics
- Trainer-Guided Integrated Project: Understanding Tokenization & Context Limits(Estimate & Optimize Tokens)
- Trainer-Guided Integrated Project: Prompt Optimization via /ai/chat/with-usage
- Trainer-Guided Integrated Project: Generate Edge Cases
- Trainer-Guided Integrated Project: From User Story to Feature File (AI-Driven BDD Generation)
- Trainer-Guided Integrated Project: Generate Locators for SauceDemo Login (Dynamic Locator Generation)
- Trainer-Guided Integrated Project: AI-Augmented Framework
- Trainer-Guided Integrated Project: Prompt Engineering Game

## Project – Hackathon

**Deliver your project and get the review comments from SME.**

The deliverables of the Hackathon will be evaluated by the BU SME.  
Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Day 63 - 65

### Evaluation:

#### Final Project Evaluation + Final Technical Evaluation

The overall Final Technical evaluation will cover topics specified in the curriculum and will be conducted through a video interview on the GenC Learn Platform.

Trainees are given two attempts to pass the final evaluation. If a Trainee receives a RAG status of “RED” or “AMBER” status on their first attempt, they have one opportunity to improve and achieve a “GREEN” status on their second attempt.

## What is Final Evaluation?

The Final Evaluation will be conducted to certify whether a GenC is eligible to enter into the BU or not. The skill of a GenC will be gauged on the application development and overall technical knowhow towards the end of GenC Training.

Tech SME from BU will be conducting the final tech evaluation. As a fallback, the project mentor can also steer this activity.

The final evaluation will be conducted as two phases. They are the following

- 1. Final Technical Evaluation**
- 2. Final Project Evaluation**

The mode of these evaluations will be any one of the following:

- F2F(face to face)
- Video Based

### 1. Final Technical Evaluation (FTE)

The BU Mentor will interview the GenC on various skills achieved throughout the training program and put a score which will be considered for the final PHS of the GenC.

### 2. Final Project Evaluation (FPE)

In this evaluation, the BU Mentor will be verifying the skills of a GenC on a project perspective. End of this evaluation, the BU Mentor will score the GenC's work based on various evaluation criterions.



## How to learn each day?

Each day has a set of learning objectives. These learning objectives can be met by going through the Udemy courses and by completing the hands-on exercises mentioned in the daily plan.

The below strategies will help you decide the learning approach.

# Learning Strategy & Approach

Find below few imaginary profiles. For each of these profiles we have defined a recommended learning approach. This is not an exhaustive list. The approaches below might help invent a new way of learning.

## Profile #1



### Harry Reacher

**Engineering Discipline:** Electronics

**Skills:** Python, Ruby on Rails, nginx

**Project:** Mining Crime Data to get Route Cause Insights

**Learning Approach to Programming Languages:** I do not want to waste my time learning. I am more practice oriented. I want to work on the problem immediately

#### What will work for me?

- Directly complete hands on exercises
- Refer Internet or Udemy Courses
- If hands on are implemented early, clarify your friends questions and troubleshoot their issues

## Profile #2



### Olivia Richards

**Engineering Discipline:** Computer Science

**Skills:** Java, C, C++

**Project:** Library Management System

**Learning Approach to Programming Languages:** I have interest, but I don't know where to start.

#### What will work for me?

- Go through the recommended Udemy Course
- Try completing the hands on exercises
- Get your clarifications solved with help from Tech SME
- Get help from other learners in your batch whom had already completed

## Profile #3



### Greg Anderson

**Engineering Discipline:** Civil

**Skills:** C

**Project:** Fiber reinforced concrete

**Learning Approach to Programming Languages:** I am scared of programming languages. I haven't got my hands dirty with coding

#### What will work for me?

- Go through the recommended Udemy Course
- Implement the coding along with the author of the Udemy Course
- Try completing the hands on exercises
- Clarify queries with SME

## FAQ

1. Who can participate in this program?

Students who have enrolled for Full Internship Program (or) the Cognizant on-boarded GEN Cs can participate in this program.

2. Is there any pre-learning I should do?

No. This program is open to all students from any academic discipline.

3. How will I know my RAG status?

It will be shown to you in the GEN C learn Platform, in your Home Page.

4. Whom do I reach out in case of any queries?

Coach is your point of contact.

5. What is the significance of Hands-on in the overall learning journey?

Hands-on focuses on specific topics in a Skill, which you can try and execute in the Platform. Group of such Hands-on exercises will be packaged together as a Code Challenge. This Code Challenge will allow you to benchmark your skills in the learning journey. Hands On/ Code Challenges/ ICT are learning components which will help you in understanding the skills better.

6. What is Code Challenge?

A problem statement will be provided to you and you need to solve it using a single skill.

7. What is Integrated Capability Test (ICT)?

A case study problem statement will be provided to you, that you may need solve using the combination of Skills learnt in the given stage.

8. How many attempts are provided for the Coding challenge and ICTs? Is it open all the time for practice?

The Coding challenges and ICTs are open and there are 3 attempts to take them up.

9. What is the entry criteria for qualifier?

A minimum of 70% hands-on completion and attempt in the CC & ICT is the eligibility criteria for qualifier.

10. What skills are covered in the qualifier?  
The skills of Stage 1 are covered in the qualifier. Only ONE attempt is provided to clear with a minimum score of 70%
11. What if I fail in the Interim evaluation?  
Your coach will notify your performance in the Interim evaluation. However you can continue with the learning.
12. How many chances will I get in the Final evaluation?  
You'll get 2 chances in the Final evaluation which covers ALL the skills in the learning journey.
13. Whom do I reach out in case of any queries?  
Coach is your point of contact.